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Designing Advanced Separators Toward Lithium-Ion Batteries

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With the deep commercialization of lithium-ion batteries (LIBs), the market has put forward higher requirements for their energy density, safety, and lifespan. The separator, as a key component of the physical barrier that hinders short circuits, offers numerous opportunities for enhancing cell performance by compensating the drawbacks of the cathode/anode or providing innovative functions. To date, considerable efforts have been devoted to the exploration of advanced separators, aiming at overcoming the challenges faced by LIBs. This review first outlines the fundamental requirements and preparation techniques of separators to guide the meticulous design. Then, the critical roles of the separator in the cathode and anode are summarized, accompanied by up-to-date progress, challenges, and potential improvement strategies. Additionally, this work emphasizes the roles of the separator in achieving high thermal safety performance in batteries and classifies its functions. Finally, an outlook on future research directions is provided to inspire the design and commercialization of separators. This review aims to deepen the understanding of the roles of separators and foster the development of separator-derived strategies for addressing issues in the field of energy storage.

1. Introduction

The escalating environmental issues have catalyzed the transformation of energy structures and the development of clean energy technologies.[1] Within these advancements, batteries stand out as a promising candidate capable of providing reliable and feasible solutions.[2] Currently, lithium-ion batteries (LIBs), as the mainstream technology in the energy storage market, are extensively applied in portable electronics, electric vehicles, and large-scale power systems.[3] Nonetheless, the development of advanced LIBs is hindered by some challenges, including insufficient cycling stability, limited energy density, and safety hazards. For instance, intercalation-type cathodes (e.g., $LiCoO_2$, $LiFePO_4$, and $LiNi_xCo_yMn_{1-x-y}O_2$) suffer from several issues, including the dissolution of metal ions, HF/H_2O attack, and lithium loss, while conversion-type cathodes (e.g., sulfur) face severe capacity fading from polysulfide shuttling phenomenon and sluggish reaction kinetics.[4] For the anode side, the dendrite evolution, a potential threat to the short circuit, requires to be well settled.[5] Moreover, in practical applications, it is also crucial to ensure high-thermal-safety performance under long-term operation or thermal runaway conditions.[6] These concerns are closely tied to the components, including the cathode, anode, electrolyte, separator, and the overall configuration of the battery. Therefore, optimizing these essential components has emerged as a pivotal focus for innovation within the battery industry.

The separator is a critical element of batteries, which does not only separate the cathode and anode to prevent short circuits but also provide storage space for the electrolyte.[7] Generally, the separator does not engage in electrochemical reactions. However, its direct contact with the cathode/anode and the electrolyte greatly affects the battery performance, including energy density, cycle life, and safety.[8] Currently, commercially available separators for LIBs are primarily made from polyolefin materials, of which polyethylene (PE) and polypropylene (PP) are the most widely used matrix materials. These materials balance excellent mechanical properties, electrochemical stability, and affordability. However, their inherently weak thermal stability introduces safety hazards, and their poor compatibility with carbonate-based electrolytes significantly reduces ionic conductivity.[9]

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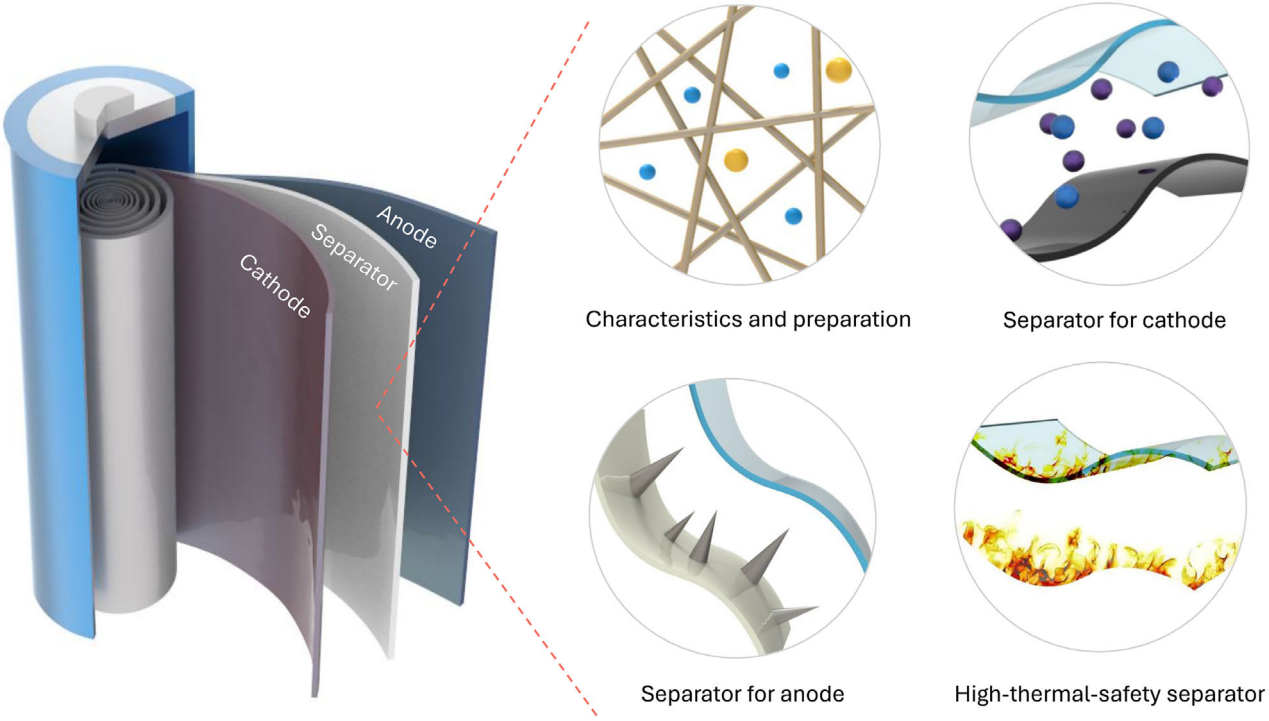


Figure 1. Schematic illustration of the main research directions of separators in lithium-ion batteries (LIBs).

Furthermore, the insufficient interfacial compatibility between polyolefin separators and the electrodes may result in the formation of unstable interfacial layers, exacerbating battery failure.[10] As a result, multiple efforts are focused on refining current commercial separators as well as developing innovative alternatives. For instance, nonwoven mats, organic/inorganic modified membranes, and electrolyte membranes have been widely explored to satisfy specific requirements.[11] This expansion of options paves the way for significant enhancements in battery performance.

To date, numerous studies have concentrated on functional separators to tackle the challenges associated with cathode and anode, as well as realizing high-thermal-safety features (**Figure 1**). In light of the significance of the separator, a systematic discussion of its roles is critical for improving the performance of LIBs. This review outlines the design criteria and techniques for separators, categorizes the functions of separators across the cathode and anode of LIBs, and elaborates on their operational mechanisms. Additionally, the effects of separators on battery safety are classified. Finally, the challenges faced by separators as critical components in practical applications and feasible solutions are proposed. This review aims to inspire researchers to engage in deeper contemplation regarding the design of separators, ultimately advancing the progress of high-performance LIBs.

2. Characteristics and Preparation Techniques of Separators

2.1. Characteristics of Separators

As a material that absorbs electrolytes and separates the cathode and anode in a battery, the primary requirement for the separator is to maintain inertness during the electrochemical reactions to prevent short circuits, while also ensuring the rapid lithium ions (Li^+) transport. In this section, we will analyze the critical parameters according to the separator structure and separator physical chemistry (**Figure 2**).

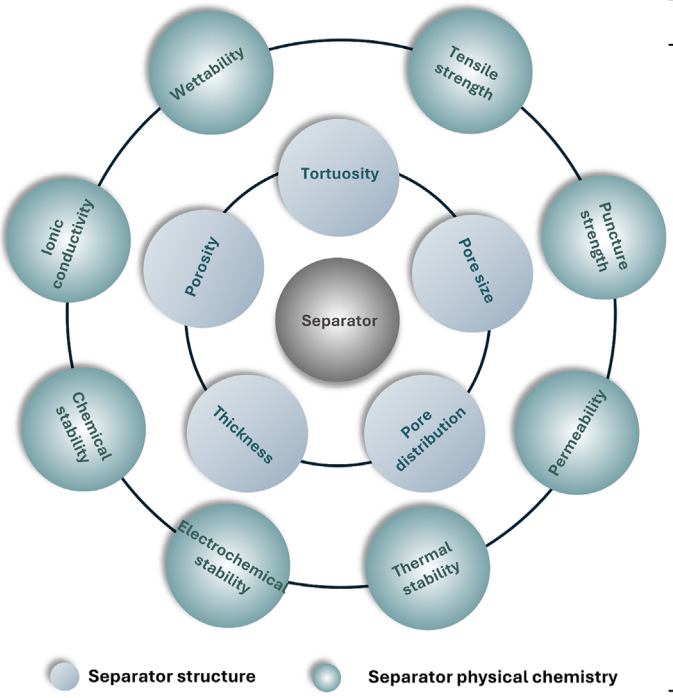
2.1.1. Separator Structure

Pore Structure: The pore structure of the separator allows it to effectively absorb the electrolyte and ensure optimal ionic conductivity, which primarily involves controlling the pore size, pore distribution, and porosity, respectively.[12] Firstly, the pore size is critically important. Smaller pores can increase the resistance to ion transport, while larger pores may lead to direct contact between electrode materials and short circuits. Typically, separators are designed with sub-micron scale pore sizes (<1 μm) to effectively block electrode particles while allowing ionic transport. Subsequently, homogeneous pore distribution is critical to maintaining uniform current density distribution, thereby suppressing localized dendrite formation and enhancing overall cell performance and cycle life. Furthermore, porosity (ε), defined as the ratio of void volume to geometric volume, serves as a key parameter that directly influences ionic conductivity and electrolyte retention capacity.[13] Low porosity hinders electrolyte absorption and increases internal resistance, whereas excessively high porosity can adversely affect the mechanical strength of the separator. An ideal porosity for separators is usually controlled between 40% and 60%.

Tortuosity: Tortuosity (τ) is a dimensionless quantity describing the influence of the morphology of the solid phase on the ion

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Parameter	Requirement
Pore size	<1 μm
Porosity	40-60 %
Pore distribution	Uniform
Tortuosity	~2.5
Thickness	20-25 μm
Permeability	<0.025 s μm^{-1}
Wettability	Wet out quickly and completely
Ionic conductivity	10^{-3} -10^{-1} S cm^{-1} (with electrolyte)
Chemical/electrochemical stability	Stable for a long period of time
Thermal stability	<5% shrinkage after 60 min/90 °C
Tensile strength	1000 kg cm^{-2} (98.06 MPa)
Puncture strength	300 g

Figure 2. Critical factors of separators, including structure and physical chemistry.

flow. By combining the information on porosity and tortuosity, the effective diffusion coefficient (D_{eff}) of Li^+ through the separator can be determined as $D_{eff} = (\varepsilon/\tau) \times D_{el}$, while the effective conductivity (σ_{eff}) can be calculated as $\sigma_{eff} = (\varepsilon/\tau) \times \sigma_{el}$, where D_{el} and σ_{el} are the diffusion coefficient and conductivity of Li^+ in the bulk electrolyte, respectively.[14] The term (ε/τ) is referred to the effective transport coefficient (δ). The typical value of the tortuosity is around 2.5, resulting in an effective transport coefficient of approximately 0.16 for the separator.

Thickness: The thickness of the separator is a key factor in balancing safety and energy density. Thicker separators can result in increased ionic resistance and occupy significant space within the battery, ultimately reducing the energy density. Conversely, thinner separators may reduce mechanical strength and heighten the risk of short circuits inside the battery. Typically, a uniform thickness of approximately 20–25 μm is preferred.[15]

2.1.2. Separator Physical Chemistry

Permeability: Permeability (κ) is also used to describe the effect of the separator's structure on ion transport, typically represented by the Gurley value, which is the time required for a certain volume of air to pass through a specified area of the separator under a certain pressure.[16] The Gurley value is closely related to porosity, pore size, thickness, and the tortuosity of the pores. The greater the air permeability of the separator, the higher the internal resistance of LIBs, which is detrimental to battery kinetics. According to the Japanese Industry Standard (JIS), the Gurley value is usually controlled between 200 and 600 s.

Ionic Conductivity: The ionic conductivity for a separator containing electrolyte is typically expected to exceed 10^{-3} S cm^{-1}. Generally, the ionic conductivity within an assembled cell can be estimated using the MacMullin number.[17] It can be calculated from the ratio of the separator resistance of the saturated electrolyte to the separator resistance of the electrolyte alone. A lower MacMullin number is promising to achieve superior cell performance.

Electrochemical/Chemical Stability: The separator acts not only as a medium for the electrolyte but also comes into direct contact with both electrodes, making its chemical stability paramount to prevent degradation from the electrolyte (e.g., ethers and esters) or electrode materials. Furthermore, electrode potentials dynamically shift during electrochemical cycling, necessitating a wide electrochemical window for the separator to avoid oxidation or reduction.[18] Therefore, when designing new membrane materials, it is essential to evaluate their lowest unoccupied molecular orbital and highest occupied molecular orbital energy levels to ensure proper electrochemical stability.

Wettability: Various molecules (e.g., anions, cations, solvent molecules) in the electrolyte are able to interact with the functional groups on the surface of the separator. These interactions are influenced by the degree of dissociation of the solvation shell around the electrolyte's salt and ions. Wettability is an essential factor in assessing the separator–electrolyte interactions, which is influenced by the surface free energy of both the separator (γ_{se}) and the electrolyte (γ_{el}).[19] Generally, wettability is evaluated by analyzing the dynamic contact angle of the separator. A smaller contact angle, along with the absorption rate of the electrolyte into the separator suggests a positive interaction between the separator and electrolyte. Associated with the wettability, the liquid retention capacity of the separator is important to ensure electrolyte storage, which can be obtained by measuring the quantity

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of electrolyte retained after submerging the separator in the liquid electrolyte for a duration of 24 h.

Mechanical Properties: The mechanical properties of the separator are crucial to the safe operation of the battery.[20] Specifically, mechanical properties are characterized by two key parameters: tensile strength and puncture strength. It is essential for the separator to possess sufficient tensile strength to withstand the forces exerted by other components. According to the American Society for Testing and Materials (ASTM) D882 and D638 standards, a separator with a thickness of 25 µm requires a minimum tensile strength of 1000 kg cm^{-2} or 98.06 MPa. In addition, the separator also needs to have high puncture strength to prevent short circuits caused by the expansion of electrode materials and the growth of lithium dendrites during cell operation. Following the ASTM D3763-18 standard, the puncture strength of a 25 µm thick separator should exceed 300 g.

Thermal Stability: In the case of a short circuit or overcharging, a substantial amount of heat is generated, which can lead to heat accumulation and rising temperatures.[21] Therefore, it is essential for the separator to maintain its integrity at elevated temperatures to ensure its functionality and prevent thermal runaway caused by the above issues. Typically, the separator should exhibit a shrinkage rate of less than 5% after 60 min at 90 °C.[22]

2.2. Preparation Techniques of Separators

The manufacturing techniques of separators are of great significance in determining the critical parameters mentioned above, ultimately affecting battery performance. This section will provide a comprehensive overview of preparation methods for separators.

2.2.1. Dry and Wet Process

Currently, the predominant production methods for polyolefin membranes in the commercial market are categorized into dry and wet processes (**Figure 3a**).[23] The dry process begins by melting polyolefin raw materials at high temperatures, followed by extrusion and blow molding to create crystalline polymers. After crystallization and thermal treatment, stretching is applied to release the crystalline layers, resulting in a porous structure. This method stands out for its simplicity and short production time. In contrast, the wet process involves combining high-boiling hydrocarbons or small molecular substances with polyolefin resin to act as pore-forming agents. The mixture is heated until it melts, allowing for phase separation upon cooling, then pressed into sheets and stretched either longitudinally or biaxially. Finally, the pore-forming agents are removed through volatile solvent extraction or evaporation. Separators produced by this technique feature high porosity and excellent gas permeability, making them well-suited for the rapid charge and discharge demands of batteries. However, both methods may struggle to remain stable under high temperatures, raising concerns about short-circuit and thermal runaway risks. Furthermore, polyolefin membranes often exhibit poor electrolyte wetting and low Li^+ conductivity, which directly hampers the overall performance of batteries.

2.2.2. Electrospinning

Electrospinning is a technique that uses a high electric field environment to produce polymer fibers with diameters typically in the range of 40 nm to 2 µm (Figure 3b).[24] The process entails the storage of a polymer solution in a capillary or syringe while applying a smoothly enhanced electric field. Once the electric field strength reaches a critical value, the surface tension of the solution is overcome by the electrostatic repulsion and forms a Taylor cone at the tip of the capillary. The polymer solution is then ejected from the tip of the Taylor cone and deposited onto a collector, where the electric field regulates the trajectory of the spraying solution. The solvent rapidly evaporates, resulting in a random distribution of polymer fibers on the collector, ultimately forming a fibrous membrane. The success of electrospinning is influenced by various factors such as the properties of the polymer solution (e.g., viscosity, surface tension, and conductivity) and operational parameters (e.g., electric field strength during spinning, distance between the needle tip and collector, etc.). Any modification to these factors can significantly alter the morphology and performance of the fibrous membrane.[25] Thus, meticulous control of these parameters is crucial for realizing the desired outcome. So far, a number of nanofiber membranes, including polyacrylonitrile (PAN),[26] polyvinylidene fluoride (PVDF),[27] and polyethylene oxide (PEO),[28] have been developed via electrospinning. By controlling the precursor, needle type, temperature and other necessary parameters, nanofibers with various structures including hollow, core–shell, blending, porous and multishell features can be obtained, which endow the separators with specific functionalities.[29]

2.2.3. Phase Separation

One of the most popular methods for fabricating porous polymer separators is phase separation (Figure 3c), which encompasses thermal-induced phase separation (TIPS),[30] nonsolvent-induced phase separation (NIPS),[31] and vapor-induced phase separation (VIPS).[32] The TIPS process requires a high-boiling, low-volatility organic solvent system in which the polymer is dissolved to form a homogeneous solution at elevated temperatures. Upon quickly cooling the coated solution on a substrate, phase separation (solid-liquid and liquid-liquid) occurs. Subsequently, the solvent is extracted using an appropriate extracting agent, resulting in a porous polymer separator with a defined structure. TIPS separators are known for their excellent mechanical properties and high reproducibility. The NIPS method is frequently employed for producing asymmetric porous membranes with a broad pore size and distribution. With this technique, the polymer is dissolved in a suitable solvent at room temperature, yielding a uniform solution that is then coated onto a substrate to form a thin film. The film is subsequently immersed in a designated nonsolvent bath, known as a coagulation bath. The polymer forms microporous structures by triggering phase separation and reverse phase transition through the exchange of solvent and nonsolvent. This exchange is realized by the chemical potential gradient created by the difference in solubility of the polymer in both. The morphology of the separator can be precisely controlled by adjusting the relationships among these components,

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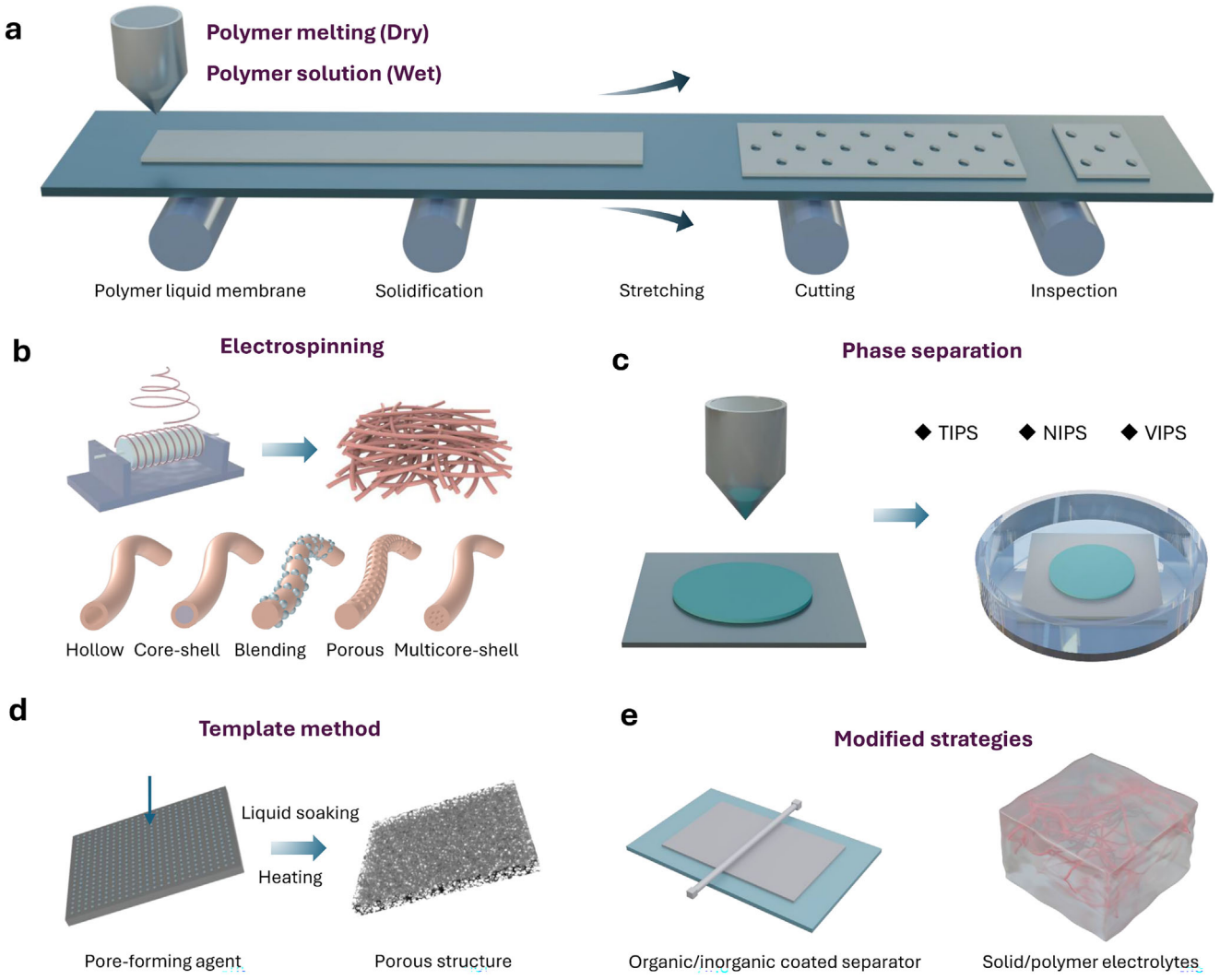


Figure 3. Preparation techniques of separator, a) dry and wet methods, b) electrospinning, c) phase separation, d) template methods, and e) innovative modified methods.

making NIPS an easy-to-operate and widely applicable technique. VIPS shares similarities with NIPS, with the key distinction being that VIPS does not require a coagulation bath. Specifically, it involves placing a thin film coated with a homogeneous solution of polymer and solvent in a vapor atmosphere composed of a nonsolvent, where the vapor permeates the solution and promotes phase separation. This technique significantly reduces the amount of organic solvent used and lowers production costs. Separators made via VIPS typically exhibit highly symmetric porous structures. By adjusting polymer concentration, the relative humidity of the vapor atmosphere and exposure time, the pore size can be tailored, allowing for a transition from symmetric to asymmetric pores.

2.2.4. Template Methods

The template method is widely recognized for its simplicity and cost-effectiveness in fabricating porous materials (Figure 3d). This process involves forming a thin film of an organic solvent containing the polymer and porous particles (e.g., water-soluble salts with specific sizes) by coating or using a three-dimensional mold. The film is then evaporated or freeze-dried to remove the solvent, and the polymer, which contains only the porous particles, is immersed in a designated solution to dissolve the particles, thereby creating a porous structure. For instance, NaCl was utilized as a sacrificial material to produce a PVDF porous membrane with larger pores measuring approximately 300–400 μm.[33] The formation of these pores takes place after polymer crystallization, in the spaces previously occupied by the sacrificial NaCl particles. It is worth noting that the size of the pores is directly influenced by the size of the sacrificial particles used. While the template technique can produce separators with highly porous structures, the template method has limitations, as it does not enable the creation of structures with fully interconnected pores that have a uniform and reproducible morphology. Therefore, the template method is rarely adopted in the preparation of commercial separators.

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2.2.5. Modified Techniques

Apart from the above techniques, to ensure that the separator achieves specific physicochemical functions in practical applications, extra design techniques such as integration of organic/inorganic materials and chemical grafting are required (Figure 3e). For example, carbon-based materials, inorganic ceramics and organic materials can be introduced into the separator via in situ or ex situ methods.[34] In addition, chemical grafting techniques can activate the inert surface of separators, creating active sites by employing pretreatments such as plasma treatment, UV radiation, and chemical initiator decomposition. Active sites help to form strong chemical bonds between the modified materials and separator, ensuring a secure and stable binding. In recent years, with the development of solid-state electrolytes, much attention has been focused on electrolyte membranes, which can simultaneously serve as an electrolyte and a separator, replacing the traditional concept of a separator.[35] For solid-state electrolytes, they are further divided into sulfide, oxide, and polymer electrolytes. Among these, sulfide electrolytes exhibit the highest ionic conductivity and are gradually becoming the main focus for the future development of all-solid-state LIBs.[36]

These methods mentioned above significantly enhance the applicability of the separator and improve the performance of the battery by acting on the cathode and anode interfaces. In the following sections, we will focus on the roles of separators for the cathode and anode of batteries, as well as their impact on the overall safety performance of the battery.

3. Separator Strategies for Cathodes

As a key component of LIBs, high-energy-density, long lifespan, and high-safety cathode materials are highly favored. Currently, intercalation-type and conversion-type cathodes are two major research directions. Among them, intercalation-type cathodes mainly store energy through the insertion and extraction of Li^+, while conversion-type electrodes undergo chemical bond breaking and formation during the electrochemical process. Although the stability of intercalation-type cathodes is much higher, the capacity is limited by theoretical constraints.[37] By contrast, conversion-type electrodes have higher specific capacities and energy densities, but the practical application is hindered by challenges including volume change, poor reversibility, and unsatisfactory cycling stability. In this section, we highlight the contributions of separators in addressing these challenges associated with the cathode, along with the typical strategies.

3.1. Separator Strategies for Intercalation-Type Cathodes

Common intercalated-type cathode materials used in LIBs mainly include layered oxides and olivine-structured compounds, such as $LiCoO_2$, $LiFePO_4$ (LFP), and $LiNi_xCo_yMn_{1-x-y}O_2$. However, these materials undergo structural changes including lattice expansion, contraction, or phase transitions during the electrochemical process, which triggers stress accumulation and particle rupture, thus adversely affecting the cycle performance and capacity retention.[38] In addition, interfacial side reactions may occur between cathode materials and electrolytes under high-pressure conditions, resulting in the loss of active substances and the leaching of transition metal ions, which in turn causes a series of chain reactions, such as the decomposition of electrolytes and the increase of interfacial impedance.[39] These structural degradations and interfacial instabilities ultimately lead to capacity degradation, shorter cycle life, and lower Coulombic efficiency. Separator strategies to tackle these challenges are categorized into three aspects: i) lithium supplement, ii) HF/H_2O elimination, and iii) solute ion adsorption.

3.1.1. Prelithiation

During the first charging process of LIBs, the electrolyte is reduced and decomposed on the surface of the anode, forming a solid electrolyte interphase (SEI), which permanently consumes Li^+ from the cathode and electrolyte, resulting in low Coulombic efficiency in the initial cycle.[40] In addition, cathode material detachment and irreversible Li^+ intercalation/deintercalation also reduce the utilization of Li. The most widely used graphite anode has an irreversible capacity loss of more than 6%, and for silicon-based and tin-based alloy anodes with high-specific capacities, the irreversible capacity loss can be as high as 10–30%.[41]

Prelithiation has been considered as an effective technology to address this problem, which involves adding a lithium source to the battery to supplement Li^+ before operation. By prelithiation, it offsets irreversible Li^+ loss, thereby increasing the total capacity and energy density of the battery. To date, a variety of cathode prelithiation strategies have emerged. For example, previous studies have explored the integration of sacrificial lithium salts (e.g., Li_2O,[42] Li_2O_2,[43] Li_2S,[44] Li_5FeO_4,[45] Li_2MoO_3,[46] $Li_6C_OO_4$,[47] and $Li_2C_2O_4$[48]) with the cathode to supplement lithium can realize an elevated Coulombic efficiency at the initial cycle. However, the decomposition of these lithium-rich additives may create poorly conductive residues or voids, and produce large amounts of gas, which hinder charge transfer and lead to degradation of the cathode structure.

Compared to conventional cathode prelithiation methods, the separator-derived strategy can effectively overcome the direct impact of prelithiation agents on the cathode, thereby improving the cycle life of batteries.[49] Generally, the prelithiation technology for separators mainly involves loading Li-rich agents onto the separator and releasing them into the battery system during the electrochemical process. An optimal prelithiation additive should provide multiple delithiation redox platforms, managing its decomposition process via voltage to ensure a controlled release of active Li^+. Additionally, the decomposition products of the additive form a protective layer that prevents dendrite penetration and absorbs excess Li^+ during overdischarge. Importantly, the products resulting from the complete or partial decomposition of the additive exhibit a suitable lithiation potential, falling between the cathode's insertion and overlithiation potentials (**Figure 4a**). In the event of overdischarge or direct contact with lithium dendrites, the lithiation of the additive occurs before that of the cathode, thereby safeguarding the cathode.

For instance, $Li_2C_4O_4$-CNT composites were coated on the separator as a sacrificial agent, gradually releasing active lithium during the battery charging process, not only compensating for the capacity loss in the first cycle but also ensuring the capacity

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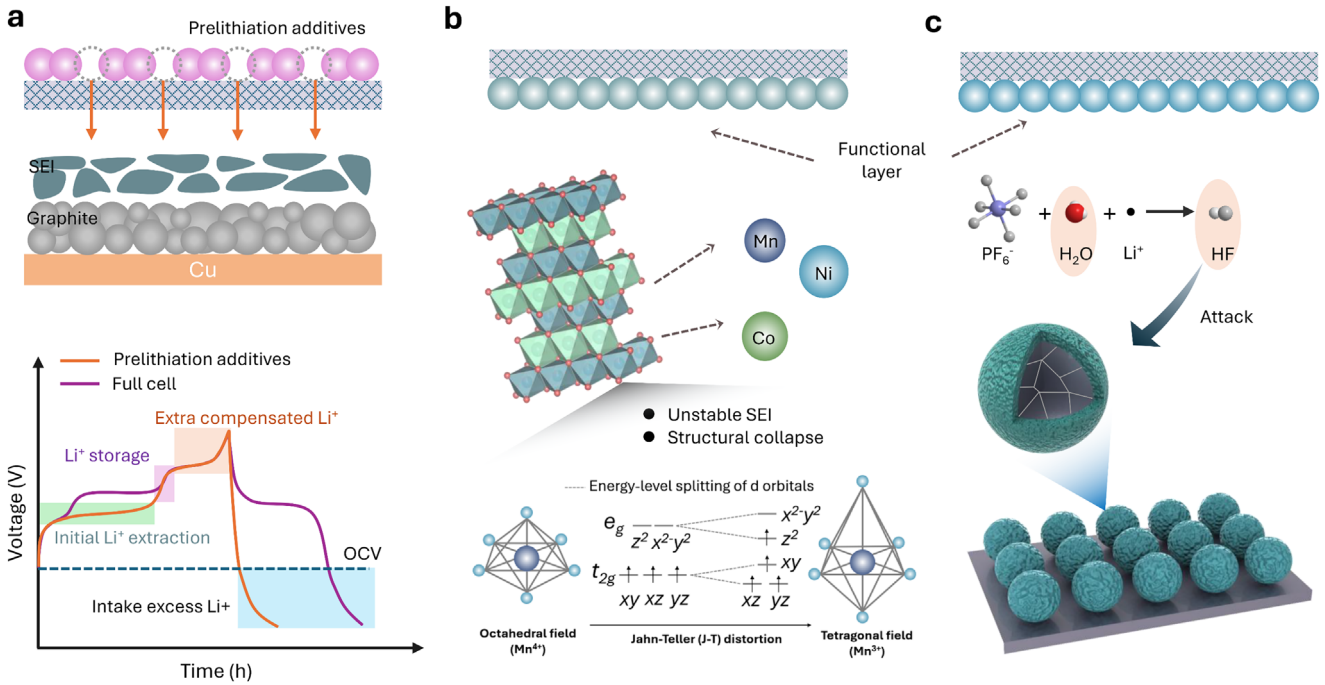


Figure 4. Separator strategies for intercalation-type cathodes, a) prelithiation, b) solute ion adsorption, and c) HF/H_2O elimination.

retention rate of the battery throughout its whole life.[50] Compared to the inorganic Li_3N and Li_3P agents, $Li_2C_4O_4$ exhibits superior air stability and is suitable for scalable production. Critically, when implemented via separator modification rather than direct cathode incorporation, its decomposition into CO_2 and C is effectively prevented by minimizing the interaction with catalytic delithiated LFP particles. This strategic placement simultaneously reduces carbon residue accumulation and enhances electrochemical stability. The lithium replenishment capacity can be precisely controlled through adjusting the charge cutoff voltage, leveraging the higher oxidation potential of $Li_2C_4O_4$ (≈3.9 V) than LFP (≈3.5 V). However, gas evolution remains an inherent challenge for overall cells during lithium compensation, as evidenced by the decomposition pathway:

$$Li_2C_4O_4 \rightarrow 4CO\uparrow + 2Li^+ + 2e^- \quad (1)$$

This necessitates a specialized battery design with an integrated gas-release valve during the formation period. On the other hand, the evolved gas would react with the electrolyte and Li^+, generating Li_2CO_3 in the cathode–electrolyte interphase (CEI). Similarly, antifluorite Li_5FeO_4 (LFO) was selected as the prelithiation agent, as it has a theoretical capacity of up to 867 mAh g^{-1}. During the initial formation process between 3.5–4.5 V, the LFO delivers a high irreversible charge capacity of 701 mAh g^{-1} due to the following four-electron redox reaction:

$$Li_5FeO_4 \rightarrow LiFeO_2 + O_2\uparrow + 4Li^+ + 4e^- \quad (2)$$

The optimized separator endows the NCM811|SiO_x/G full battery with an increased reversible capacity of 40%.[51] Notably, the separator prevents Fe cross-over from the cathode side to the anode SEI, ensuring the interfacial stability. The gas evolution is also negligible in contrast to $Li_2C_4O_4$. Nevertheless, LFO is vulnerable to air exposure and requires further coating engineering using Li_6CoO_4 or a careful treatment under an inert atmosphere.

The prelithiation technology can also be used to activate recycled spent cathodes, repairing their structure and elements. For example, surface-engineered Li_5FeO_4 (SE-LFO) was utilized as a prelithiation agent coated on the cathode side of the separator.[52] At the initial formation stage, the SE-LFO selectively releases a portion of its capacity, which can regenerate spent LFP and prolong its life. During the service phase, the remaining capacity of the SE-LFO can be reactivated through potential regulation, compensating for the loss of active lithium that causes cathode capacity degradation during long-term cycling, thereby extending the battery's cycle life. Additionally, the decomposition products of SE-LFO have been proven to act as a protective layer, preventing the cathode from being overlithiated under overdischarge conditions, which can lead to electrochemical failure. This significantly enhances the battery's safety and sustainability.

3.1.2. Solute Ion Adsorption

Attributing to the Jahn–Teller effect, electrolyte oxidation, cathode phase transition, and corrosion by acidic substances during the electrochemical process, transition metal ions in the cathode material will initially dissolve into the electrolyte, manifesting as abnormal capacity performance and voltage polarization.[53] The dissolved metal ions form free transition metal complexes in the electrolyte and tend to deposit on the surface of the anode. Over time, the deposited transition metals will continuously catalyze the interface reaction between the anode and the electrolyte, exacerbating the generation and decomposition-regrowth of the SEI, ultimately leading to capacity decay.[54] In addition, these

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interrelated reaction processes often involve the production and consumption of gases. Studies have shown that Ni, Co, and Mn can all dissolve from the cathode, whereas Mn causes the greatest damage to the anode SEI.[55]

Anchoring transition metal ions through a separator can effectively alleviate the impact of dissolved ions on the anode (Figure 4b). To date, porous materials like metal organic frameworks (MOFs) and covalent organic frameworks (COFs) with large specific areas and active sites have been proved to be able to effectively promote Li^+ transport and absorb the transition metal ions. For example, Zhang's group used chemical modification techniques to graft ethylene diamine tetraacetic acid (EDTA) molecules onto MOF-808 to design a multifunctional ion-selective separator.[56] The constructed separator provides uniform capture sites for transition metal ions and fast Li^+ transport channels to prevent crossover between the cathode and the anode. The negative electronegativity of the N-/O- groups in EDTA molecules is relatively strong, which shows a strong chemical interaction with metal ions. Under harsh test conditions such as high voltage and high temperature, this modified separator exhibits excellent electrochemical performance. Analogously, Sun's group modified different thicknesses of hydrogen-bonded organic framework material (HOF) cyanuric acid melamine (MCA) functional layers on a PP separator.[57] MCA has a pore size of 0.9 nm and can achieve selective Li^+ transport through physical size sieving. At the same time, multiple polar groups such as C = N and C = O in MCA exhibit strong adsorption with the transition metal elements dissolved from NCM, preventing the shuttling effect of transition metal elements produced by high-nickel ternary cathode during cycling.

The formation of CEI during electrochemical reactions is crucial for maintaining the stability of the cathode. Guo's group proposed a carefully designed tandem separator (CYANO-COF|PP|SnF_2) to regulate the chemical stability of CEI.[58] The cyanide groups in CYANO-COF induce the formation of a stable, CN-rich CEI on the surface of the NCM811 cathode through adsorption/coordination effects with transition metals, suppressing irreversible phase transitions, TM dissolution, and other side reactions. Simultaneously, an artificial mixed SEI layer consisting of LiF and Li-Sn alloy was constructed by a simple in situ conversion reaction, and dendrite-free Li deposition was achieved at the SEI/Li interface. Under the synergistic effect of the dual interfaces, the NCM811|Li battery maintains a capacity retention rate of 81.8% after 200 cycles at 4.5 V. However, most of the above strategies focus on porous organic/covalent framework materials, and there is an urgent need to develop efficient and low-cost alternatives.

3.1.3. Eliminating HF/H_2O Molecules

The liquid electrolyte is composed of lithium salts and organic solvents, which exhibit extremely high sensitivity to moisture. The presence of H_2O within the battery triggers the hydrolysis of lithium salts, producing LiF and highly corrosive HF. HF, in turn, attacks the CEI and SEI, leading to severe parasitic reactions in the electrolyte, ultimately damaging the electrochemical performance of the battery (Figure 4c). These issues are more pronounced in lithium metal batteries, as HF reacts violently with the lithium metal anode, leading to accelerated capacity fade and increased safety risks. Moreover, these adverse effects are more evident in high-voltage operating nickel-rich cathodes. The electrode and interface structural degradation caused by HF poses a significant challenge to high-voltage lithium metal batteries. Numerous works have been devoted to capturing H_2O or HF by electrolyte additives.[59] However, these methods can only convert protons into less acidic substances, which reach the electrodes with free movement during cycling, resulting in degradation of the battery performance. By contrast, separator strategies are effective in addressing these challenges.

For instance, a separator strategy, which involves decorating piperidine (PI) modified molecular sieves (TW) on a conventional polyethylene separator was proposed (TW@PI-PE).[60] The N-H bond in PI acts as an active site, capable of abstracting H^+ from HF, thereby triggering a protonation reaction and consuming HF. The porous structure of TW serves as a reaction chamber for PI and HF. Consequently, when using the TW@PI-PE separator, the HF content in the $LiPF_6$-based electrolyte with a certain amount of H_2O is significantly reduced, thereby protecting the nickel-rich cathode from HF corrosion. After undergoing 100 cycles under conditions of 1000 ppm water content and a cut-off voltage of 4.6 V, the capacity retention rate of the full cell can reach 81%. Similarly, a composite coating of MS and sulfonated melamine-formaldehyde condensate (SMF) is constructed on a separator to capture residual HF and H_2O.[61] With the electrolyte consisting of 1 M $LiPF_6$ in EC:DEC:EMC with 62.4 ppm H_2O, the NCM811|Li battery assembled with an MS|PE separator has a capacity retention rate of 86.8% after 250 cycles, which is significantly higher than the battery assembled with a PE separator (68.2%).

Apart from modified molecular sieves, MOF materials (e.g., MIL-101 (Cr)) were utilized to effectively and selectively capture protonic impurities in the electrolyte, achieving water-resistant and HF-resistant lithium batteries.[62] MIL-101 exhibits a unique proton affinity due to the isolated ionic fluorine sites, which can effectively capture ppm-level H_2O/HF from highly polar electrolyte solvents. The $LiNi_{0.6}Mn_{0.2}Co_{0.2}O_2$|Li battery with MIL-101 modified separator achieved over 300 cycles in the presence of 800 ppm H_2O. Compared to batteries using traditional standard separators, this battery's lifespan is extended by 10 times, demonstrating its excellent electrochemical cycling performance.

3.2. Separator Strategies for Conversion-Type Cathodes

The most commonly used conversion-type cathodes in lithium batteries are halogen electrodes.[63] Taking sulfur as a typical example, it offers significant advantages, including a high theoretical capacity (1675 mAh g^{-1}) and energy density (2600 Wh kg^{-1}). This represents a potential 3–5 times increase in energy density compared to commercial intercalation-type cathodes.[64] Additionally, sulfur is environmentally friendly, naturally abundant, and cost-effective. Therefore, Li-S batteries are considered one of the next-generation secondary energy storage systems. The conversion reaction mechanism of Li-S batteries involves solid-state redox reactions, the breaking and recombination of S-S bonds, and multielectron transfer electrochemistry. As illustrated in **Figure 5a**, the typical composition, existing form, phase

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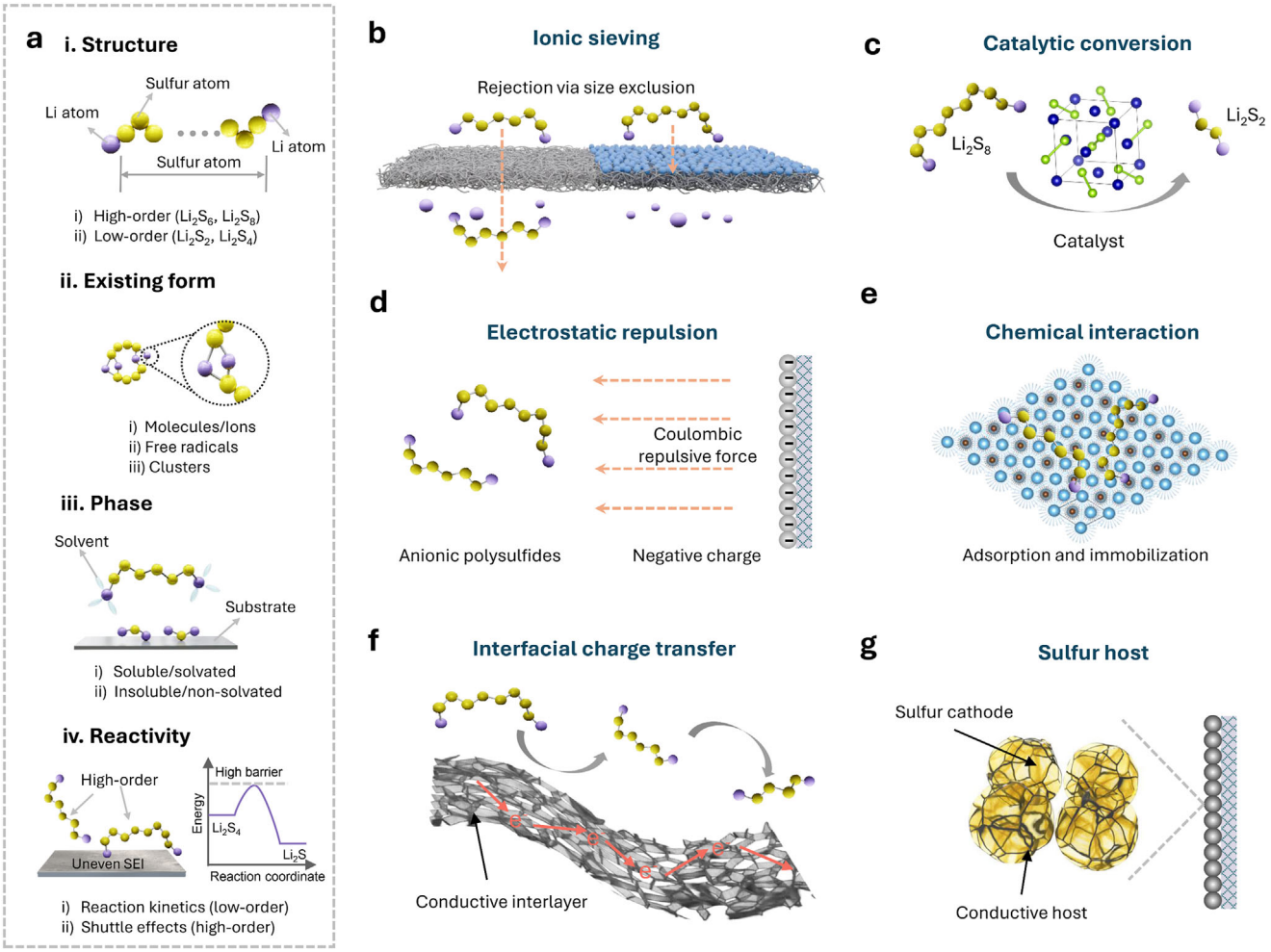


Figure 5. a) The molecular-scale properties of lithium polysulfides (LiPSs) in Li-S batteries, and b–g) schematic illustration of the functions of separators for sulfur cathodes.

and reactivity of lithium polysulfides (LiPSs) in the Li-S system are demonstrated. During the whole process, Li-S batteries face significant challenges in practical applications, including poor kinetics and the polysulfide shuttle effect. These challenges result in poor cycling stability, rapid capacity decay, low coulombic efficiency, and low sulfur utilization, impeding the commercialization of Li-S batteries.[65] The roles of separators in solving these problems are detailed below.

3.2.1. Ionic Sieving

Ionic sieving is achieved by size exclusion between long-chain polysulfides and the small nanopores in the separator (Figure 5b). In addition, separators with complex pore systems, such as hierarchical pore structures, are also effective in trapping LiPSs. A direct method is to coat the separator with functional layers that physically block the passage of polysulfides while allowing Li^+ to diffuse. Computational studies have shown that the radii of Li_2S_4, Li_2S_6, and Li_2S_8 are 5.71, 6.60, and 7.21 Å, respectively,[66] while the Stokes radius of Li^+ in propylene carbonate solvent is 4.8 Å.[67] Therefore, precise control of the pore size in the functional layers can achieve effective physical confinement. For example, materials with specific sizes, such as polypyrrole,[68] polydopamine (PDA),[69] and polyamide-66,[70] have also been used as coating materials for commercial separators and achieved some success. Yu et al. succeeded in achieving a significant narrowing of the pore size distribution by incorporating a polymeric material with a pore size less than 2 nm into the pores of a PP separator, thereby effectively blocking LiPSs.[71] Experimental validation confirmed its potential in reducing polysulfide crossover, achieving a capacity retention of 81.7% after 200 cycles at 0.1 C, compared to only 4.5% with the original separator.

Although physical confinement has a certain effect, there are inevitably some mesopores and macropores in functional materials, especially larger voids between particles. Therefore, the simple ion blocking effect is limited in suppressing the shuttling of polysulfides and often requires synergy with other effects to further enhance its effectiveness.

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3.2.2. Catalytic Conversion

A promising strategy is the introduction of electrocatalysts into the functional separator to improve electron mobility, increase Li_2S nucleation, and lower activation energy (Figure 5c). Transition metal compounds, such as Fe, Ni, Co, Mo-based materials, have been widely studied for their catalytic activity in the conversion of polysulfides. These materials can effectively promote the redox reactions of polysulfides, enhancing the overall performance and stability of Li-S batteries. By facilitating the conversion of higher-order polysulfides to lower-order ones and vice versa, these transition metals help to mitigate the shuttle effect, improve the utilization of sulfur, and enhance the battery's cycle life and rate performance. For instance, MoO_2-Mo_2N,[72] C_3N_4-$CoSe_2$,[73] FeS_2@TiO_2,[74] and Co_9S_8/CoO[75] have been proved to improve the redox reactions of the cathode. Given that the electrocatalytic performance of insulating inorganic materials for high sulfur utilization is often unsatisfactory, in some cases, conductive networks, such as carbon materials, MXene, and graphene, have been introduced to facilitate rapid electron transfer, such as PMo_{12}/$CoSe_2$@NC/CNTs,[76] rGO/MoS_2/C,[77] P-MoO_{2-x}/nano porous carbon,[78] Bi_2MoO_6-PPy,[79] Gd_2O_3/Co@NC,[80] CoN@NCNT,[81] and MXene@CoS_2.[82]

Catalytic sites can also be integrated into other materials. For example, Co nanoparticles (NPs) were evenly loaded on zeolitic imidazolate framework-8 (ZIF-8)-derived hollow and porous N-doped carbon frameworks (Co-HMCF) for highly active catalysts.[83] The tiny Co NPs promote the rapid conversion of soluble LiPSs and solid Li_2S, thereby enhancing the overall sulfur redox reaction. The optimized cell exhibits merely 0.047% capacity decay per cycle for 800 cycles at 1.0 C. Even under a current density of 5.0 C, a capacity of 582 mAh g^{-1} can be retained. Qu's group adopted various metals (M), including Mn, Fe, Co, Ni, and Cu as nodes to connect with 5,10,15,20-tetrakis(4-pyridyl) cobalt porphyrin (CoTPyP), obtaining a series of MOFs (CoTPyP-M).[84] Among these, CoTPyP-Mn exhibits the longest bond length of S-S than the other CoTPyP-M, indicating its weak effect on the S-S bond in Li_2S. As a result, CoTPyP-Mn delivers optimized sulfur utilization and stability.

3.2.3. Electrostatic Repulsion

This approach utilizes the repulsive forces between the functional separator and the polysulfide anions (S_x^{2-}) dissolved in the solution. This repulsion helps prevent the crossover of LiPSs (Figure 5d). Anionic groups, such as SO_3^-, COO^-, and CO^-, have been shown to repulse LiPSs and enhance the kinetics of cation diffusion in the electrolyte, leading to reduced chemical polarization and improved rate capability. For example, the Nafion-based film with anionic sulfonate-terminated perfluoroalkyl ether groups has been intensively reported as a cation permselective sieve, allowing Li^+ to pass through while rejecting polysulfides due to electrostatic repulsion.[85] In addition, various materials terminated with anionic functional groups, including acetylene black,[86] modified MOFs,[87] graphene,[88] and multi-walled carbon nanotubes (MWCNTs),[89] have also been shown to enhance the permselectivity of the separator.

Apart from anionic groups, inorganic materials, such as montmorillonite with negative charges[90] and $BaTiO_3$,[91] have also been explored to exhibit an electrostatic repulsion effect. For example, the dipoles of $BaTiO_3$ particles on the PE separator can be aligned along the direction of the pretreated electric field, delivering strong and persistent polysulfide repulsion. In contrast, for the $BaTiO_3$@PE separator without pretreatment in an electric field, extensive polysulfide shuttling still occurred, highlighting the crucial role of electrostatic repulsion.

3.2.4. Chemical Interaction

Chemical interactions, including Lewis acid-base interactions, polar–polar interactions, and catenation interactions, can independently or synergistically enhance the adsorption of functional separator materials for polysulfides (Figure 5e). The dissociated anions in the electrolyte can be considered as Lewis bases and can interact with various Lewis acids on the separator to form coordination bonds. In addition, the intermolecular forces between polar and nonpolar molecules are significantly weaker than dipole-dipole intermolecular forces within polar molecules. Therefore, polar polysulfides can be anchored by loading a polar coating on the separator.

Introducing heteroatoms (e.g., nitrogen,[92] boron,[93] phosphorus,[94] sulfur,[95] or co-doping[96]) into carbon materials can endow the separator with abundant polar or Lewis acid sites, which have been proven to effectively capture active materials. At the atomic scale, the electronegativity difference between carbon and heteroatoms can redistribute electrons, thereby increasing the polarity of carbon. For example, boron has a lower electronegativity than carbon. When boron is integrated into a carbon framework, the binding energy of B-doped sites with sulfides Li_2S_2 increases from 0.32 to 1.54 eV compared to undoped carbon. In addition, metal compounds, such as metal oxides (e.g., Al_2O_3,[97] TiO_2,[98] SnO_2,[99] and MnO_2[100]), metal phosphides (e.g., CoP,[101] MoP,[102] and MoP_2[103]) and metal sulfides (e.g., Co_9S_8,[104] ZnS_2,[105] MoS_2,[106] CoS,[107] ReS_2,[108] WS_2,[109] Sb_2S_3,[110] and VS_2[111]), have been employed as functional layers to immobilize LiPSs.

3.2.5. Interfacial Charge Transfer

To address the issues of sluggish redox kinetics, high polarization, and poor utilization of active materials caused by the inherently low electronic conductivity of sulfur cathodes and their discharge products, using the cathode side of the separator as a secondary conductive interface can effectively optimize redox kinetics (Figure 5f). Typically, the conductive interface on the separator can serve as a current collector. Some reports have shown that enhancing interfacial conductivity on the polymer separator is more effective in improving sulfur utilization and cycling stability compared to incorporating conductive materials into the cathode. Commonly used conductive coating materials for separators include conductive carbon black,[112] carbon nanofibers,[113] graphene[114] among others. At present, a considerable part of the work achieves multifunctionality by utilizing these conductive materials in a composite with other functional materials.

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3.2.6. Sulfur Hosts

Generally, the sulfur cathode is loaded in the porous material and integrated cast onto the current collector. However, after preparing a porous conductive functional layer on the surface of the separator, these interfacial conductive layers can also serve as carriers for the sulfur cathode, potentially replacing the conventional aluminum current collector and S host (Figure 5g), such as S-G@PP,[115] PP/d-Ti_3C_2,[116] GF/$Ti_3C_2T_x$/CNTs-PEI,[117] and PP + G/CNTs.[118] As an example, sulfur was loaded on a graphene film which was directly coated onto the separator (denoted as S-G@PP). The structure exhibits a much lower charge-transfer resistance (56 Ω) compared to that of a conventional cell configuration (291 Ω). Impressively, the integrated electrode demonstrates excellent flexibility, maintaining an electric conductivity with approximately 800 S m^{-1} even after 50 000 bending cycles.

The above roles of the separator and typical work are summarized in **Table 1**.

The aforementioned work primarily summarizes the classification of separator functions in sulfur cathodes. However, considering that intercalation-type cathodes face similar challenges, these strategies are equally applicable to other cathodes such as selenium, iodine and chlorine. For instance, N-doped carbon nanocage (hNCNC) coating was applied on PP separators for Li-Se cells, which effectively suppresses polyselenide shuttling through combined chemical adsorption and electrocatalytic effects.[119] For Li-I_2 cells, Xu's group employed an in situ growth method to prepare amorphous aramid fiber coupled with Fe-MOF.[120] The distorted metal nodes in the amorphous MOFs altered the electronic configuration, thereby enhancing interactions with polyiodides (e.g., I_2, LiI_3, LiI_5, etc.). This strengthened interaction effectively suppresses the shuttle effect of polyiodides in the electrolyte, mitigating capacity fading and ionic conductivity degradation caused by their dissolution. Similarly, some advanced modified separators, such as Mo-rGO@PP,[121] Co/CoN_4@KB-MXene,[122] MXenes/CNT/CTAB/PP,[123] TPB-DMTP-COF/PP,[124] have been successfully implemented in Li-Se, I_2, and Cl_2 battery systems, demonstrating broad applicability across various conversion-type cathodes.

4. Separator Strategies for Anodes

4.1. Anodes for LIBs

The types of anodes can be classified into carbon-based and noncarbon-based materials. Carbon-based materials include natural graphite, artificial graphite, mesocarbon microspheres, hard carbon, and soft carbon. Noncarbon types include silicon-based materials, titanium-based materials, tin-based materials, and metallic lithium. In addition to the limitations of the inherent properties of the material, the dendrite issue is a common challenge faced by all anode materials, which severely limits their practical applications as anodes. When the separator has poor wettability with the electrolyte and an uneven pore size distribution, it leads to an uneven Li^+ flux, which in turn causes nonuniform nucleation during the lithium reaction process and the formation of lithium dendrites.[125] As the number of battery cycles increases, the dendrites continue to grow until they penetrate the separator, causing serious safety issues such as short circuits. Based on the formation mechanism of lithium dendrites, the rational design of the separator can effectively realize a uniform deposition morphology. The functions of separators for anodes are classified in the following section.

4.2. Effects of Separators

4.2.1. Wettability

The wettability between the separator and electrolyte greatly affects the uniform transfer of Li^+.[126] The electrolyte serves as the medium for ion diffusion, while the porous structure of the separator provides channels for ion diffusion. A separator with good wettability facilitates the full penetration of the electrolyte. As shown in **Figure 6a**, when the electrolyte completely fills the pores within the separator, a uniform Li^+ transfer channel is established. Conversely, if the wettability of the electrolyte is insufficient, it can lead to incomplete pore filling and selective ion movement, ultimately resulting in nonuniform ion distribution on the lithium metal surface. Generally, the low-polarity C–H bonds on the surface of polyolefin-based separators often result in poor wettability. Therefore, regulating polar functional groups in the separator, such as -C = O, -NH_2, and -OH, can improve the wettability of polar solvents through electrostatic forces or hydrogen bonding. For example, Liang's Group used a polyimide aerogel with customized nitrogen functional sites as a functional separator for lithium metal batteries, significantly improving wettability from 72.3° to 0°.[127] Additionally, better wettability helps form a uniform electrolyte film on the separator surface, allowing Li^+ to reach the electrode more uniformly. This can reduce the local concentration gradient of Li^+, improving the uniformity of plating and stripping performance. Therefore, the wettability between the separator and electrolyte is a key factor to be considered when designing functional separators for lithium metal batteries.[128]

4.2.2. Homogeneous Pores

When Li^+ ions pass through traditional polyolefin separators, they tend to accumulate near the separator pores, leading to lithium ion-rich regions near the separator pores and lithium ion-poor regions near the separator framework. Therefore, constructing separators with uniform pore structures or modifying separators with regularly porous materials to provide uniform physical channels for Li^+ flux is considered an effective modification strategy to homogenize the ion flow (Figure 6b).

The templating method is an effective way to construct uniform pores in polymers. Porous anodic aluminum oxide (AAO) has highly ordered Li^+ diffusion channels, making it an ideal structure for battery separators with extremely low tortuosity. He et al. reported the preparation of AAO separators with uniformly sized vertical through-holes and a hexagonal packing array.[129] Compared to PP separators, the straight channels significantly reduce the tortuosity of the separator and regulate the Li^+ concentration within the channels. Additionally, separators prepared by the templating method can also exhibit uniform mesoporous structures. For example, using a poly(lactic acid)-polyimide-poly(lactic acid) (PLA-PI-PLA) triblock copolymer as a

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Table 1. The summary of the roles of the separator and typical work.

Separator	Thickness or mass loading	Cathode	Electrolyte	Stability capacity after *N* cycles	Refs.
Ionic sieving					
PP/PE/PP/PPy	15–25 nm/≈0.13 mg cm^{-2}	S/CB	1 M LiTFSI in DOL/DME+$LiNO_3$	805 (250 cycles) at 0.5 C	[68]
PP/Graphene	NA	S/CB	1 M LiTFSI in DOL/DME+$LiNO_3$	576.2 (95 cycles) at 0.2 C	[70]
PP/PIN	1.2–1.5 mg cm^{-2}	S/Super P	1 M LiTFSI in DOL/DME+$LiNO_3$	780 (200 cycles) at 0.2 C	[71]
Catalytic conversion					
PP/MoO_2-Mo_2N	9 µm/0.51 mg cm^{-2}	S/CB	1 M LiTFSI in DOL/DME+$LiNO_3$	632.6 (500 cycles) at 1 C	[72]
PP/C_3N_4-$CoSe_2$	37 µm/0.5 mg cm^{-2}	S/KB	1 M LiTFSI in DOL/DME+$LiNO_3$	715.6 (500 cycles) at 1 C	[73]
PP/FeS_2@TiO_2	0.4–0.8 mg cm^{-2}	S/KB	1 M LiTFSI in DOL/DME	442 (1000 cycles) at 1 C	[74]
PP/Co_9S_8/CoO	0.3 mg cm^{-2}	S/ graphene	1 M LiTFSI in DOL/DME+$LiNO_3$	769 (300 cycles) at 1 C	[75]
PP/PMo12/$CoSe_2$@NC/CNTs	≈8 µm	S/CNTs	0.5 M LiTFSI in DOL/DME+$LiNO_3$	635.77 (500 cycles) at 3 C	[76]
PP/rGO/MoS_2/C	NA	S/Super P	1 M LiTFSI in DOL/DME	642 (1000 cycles) at 1 C	[77]
PP/P-MoO_{2-x}/NPC	10.4 µm	S/CB	1 M LiTFSI in DOL/DME+$LiNO_3$	574 (388 cycles) at 1 C	[78]
Bi_2MoO_6-PPy@PP	12 µm/0.29 mg cm^{-2}	S/rGO	1 M LiTFSI in DOL/DME+$LiNO_3$	596 (500 cycles) at 2 C	[79]
Gd_2O_3/Co@NC-PP	15 µm	S/HPC	1 M LiTFSI in DOL/DME+$LiNO_3$	504.2 (500 cycles) at 2 C	[80]
PP/CoN@NCNT	0.6 mg cm^{-2}	S/CNTs	1 M LiTFSI in DOL/DME+$LiNO_3$	729.7 (200 cycles) at 0.2 C	[81]
PP/MCCoS	NA	S/KB	1 M LiTFSI in DOL/DME+$LiNO_3$	390 (1000 cycles) at 7 C	[82]
PP/Co-HCF	0.6–0.8 mg cm^{-2}	S/Super P	1 M LiTFSI in DOL/DME+$LiNO_3$	675 (800 cycles) at 1 C	[83]
PP/CoTPyP-Mn	0.33 mg cm^{-2}	S/CNTs	1 M LiTFSI in DOL/DME+$LiNO_3$	598.2 (1000 cycles) at 2 C	[84]
Electrostatic repulsion					
PP/PE/PP/Nafion	0.7 mg cm^{-2}	S/CNTs	1 M LiTFSI in DOL/DME	468 (500 cycles) at 1 C	[85a]
PP/GO/Nafion	≈0.13 µm/0.0532 mg cm^{-2}	S/G/CNTs	1 M LiTFSI in DOL/DME	800 (200 cycles) at 0.5 C	[85b]
PP/AB-SO_3^-	6 µm/0.13 mg cm^{-2}	S/C-PANI	1 M LiTFSI in DOL/DME+$LiNO_3$	955 (100 cycles) at 0.1 C	[86]
PVDF/UiO-66-SO_3Li	22 µm	S/CMK3	1 M LiTFSI in DOL/DME+$LiNO_3$	580 (500 cycles) at 0.5 C	[87]
PP/rGO@SL	0.2 mg cm^{-2}	S/C	1 M LiTFSI in DOL/DME+$LiNO_3$	450 (1000 cycles) at 2 C	[88]
PP/MMT	0.825 mg cm^{-2}	S/MWCNT	1 M LiTFSI in DOL/TEGDME+$LiNO_3$	924 (200 cycles) at 0.1 A g^{-1}	[90]
PE-Poled BTO	2.4 mg cm^{-2}	S/KB	1 M LiTFSI in DOL/DME+$LiNO_3$	929.5 (50 cycles) at 0.5 C	[91]
Chemical interaction					
PP/NHG	≈0.1 mg cm^{-2}	S/G	1 M LiTFSI in DOL/DME+$LiNO_3$	403 (800 cycles) at 2 C	[92]
PP/FBN/G	NA	S/Super P	1 M LiTFSI in DOL/DME+$LiNO_3$	700 (1000 cycles) at 1 C	[93]

(*Continued*)

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Table 1. (Continued)

Separator	Thickness or mass loading	Cathode	Electrolyte	Stability capacity after *N* cycles	Refs.
PP/Black phosphorus	0.7–1.4 µm/≈0.4 mg cm^{-2}	S/Super P	1 M LiTFSI in DOL/DME+$LiNO_3$	800 (100 cycles) at 0.4 A g^{-1}	[94]
PP/NS-doped mesoporous carbon	0.5–0.6 mg cm^{-2}	S/SPC	1 M LiTFSI in DOL/DME+$LiNO_3$	740 (500 cycles) at 0.5C	[95]
PP/HNPC	25 µm/0.36 mg cm^{-2}	S/AB	1 M LiTFSI in DOL/DME+$LiNO_3$	930 (200 cycles) at 0.2 C	[96]
PP/Al_2O_3	14 µm	S/CB	1 M LiTFSI in DOL/DME+$LiNO_3$	593.4 (50 cycles) at 0.2 C	[97]
PP/TiO-C65	7.5 µm/0.7 mg cm^{-2}	S/Super C65	1 M LiTFSI in DOL/DME+$LiNO_3$	533 (300 cycles) at 2 C	[98]
PP/HPC@SnO_2	0.2 mg cm^{-2}	S/HPC	1 M LiTFSI in DOL/DME+$LiNO_3$	832 (100 cycles) at 0.5 C	[99]
PP/G/M@CNT	0.05 mg cm^{-2}	S/CNT	1 M LiTFSI in DOL/DME+$LiNO_3$	293 (2500 cycles) at 1C	[100]
PP/CoP/KB	15.37 µm/0.2 mg cm^{-2}	S/rGO	1 M LiTFSI in DOL/DME+$LiNO_3$	500 (500 cycles) at 1 C	[101]
PP/MoP/rGO	10 µm/0.35–0.45 mg cm^{-2}	S-CNT-AB	1 M LiTFSI in DOL/DME+$LiNO_3$	759 (120 cycles) at 0.1C	[102]
PP/CNT/MoP_2	0.34 mg cm^{-2}	S/Super P	1 M LiTFSI in DOL/DME+$LiNO_3$	905 (100 cycles) at 0.2 C	[103]
PP/Co_9S_8	10 µm/0.16 mg cm^{-2}	S/Super P	1 M LiTFSI in DOL/DME+$LiNO_3$	820 (1500 cycles) at 1 C	[104]
PP/ZnS-SnS@NC	0.5 mg cm^{-2}	S/CNT	1 M LiTFSI in DOL/DME+$LiNO_3$	1149 (300 cycles) at 0.2 C	[105]
PP/MoS_2	350 nm	S/CB	1 M LiTFSI in DOL/DME+$LiNO_3$	401 (600 cycles) at 0.5 C	[106]
PP/CNF/CoS/KB	0.36 mg cm^{-2}	S/Super P	1 M LiTFSI in DOL/DME+$LiNO_3$	450 (760 cycles) at 1 C	[107]
PP/ReS_2@NG-CP	0.5 µm/80–90 µg cm^{-2}	S/CNT	1 M LiTFSI in DOL/DME+$LiNO_3$	1000 (100 cycles) at 0.3 C	[108]
PP/SSNS/CNT	0.5 µm/0.4 mg cm^{-2}	S/KB	1 M LiTFSI in DOL/DME+$LiNO_3$	680 (100 cycles) at 0.2 C	[110]
D-HVS@PP	0.14 mg cm^{-2}	S/Super P	1 M LiTFSI in DOL/DME+$LiNO_3$	908 (300 cycles) at 0.2 C	[111]
Interfacial charge transfer					
PP/Super P	1 mm/0.5 mg cm^{-2}	S/Super P	1 M LiTFSI in DOL/DME+$LiNO_3$	600 (200 cycles) at 0.1 C	[112]
PP/GO/CNT	30 µm/1.1 mg cm^{-2}	S/Super P	1 M LiTFSI in DOL/DME+$LiNO_3$	671 (300 cycles) at 0.2 C	[113]
PP/Cellular graphene framework	30 µm/0.3 mg cm^{-2}	S/CB	1 M LiTFSI in DOL/DME + $LiNO_3$	800 (300 cycles) at 0.5 C	[114]
Sulfur hosts					
S-G@PP	30 µm	S-G@PP	1 M LiTFSI in DOL/DME+$LiNO_3$	663 (500 cycles) at 0.1 A g^{-1}	[115]
PP/d-Ti_3C_2	≈0.4 mg cm^{-2}	a-Ti_3C_2.S/d-Ti_3C_2/PP	1 M LiTFSI in DOL/DME+$LiNO_3$	632 (50 cycles) at 0.5 C	[116]
GF/$Ti_3C_2T_x$/CNTs-PEI	0.2 mg cm^{-2}	S/$Ti_3C_2T_x$ /CNTs-PEI	1 M LiTFSI in DOL/DME + $LiNO_3$	980 (500 cycles) at 1 C	[117]
PP/G/CNTs	15.2 µm/0.2 mg cm^{-2}	S/G/CNTs	1 M LiTFSI in DOL/DME + $LiNO_3$	445 (500 cycles) at 2 C	[118]

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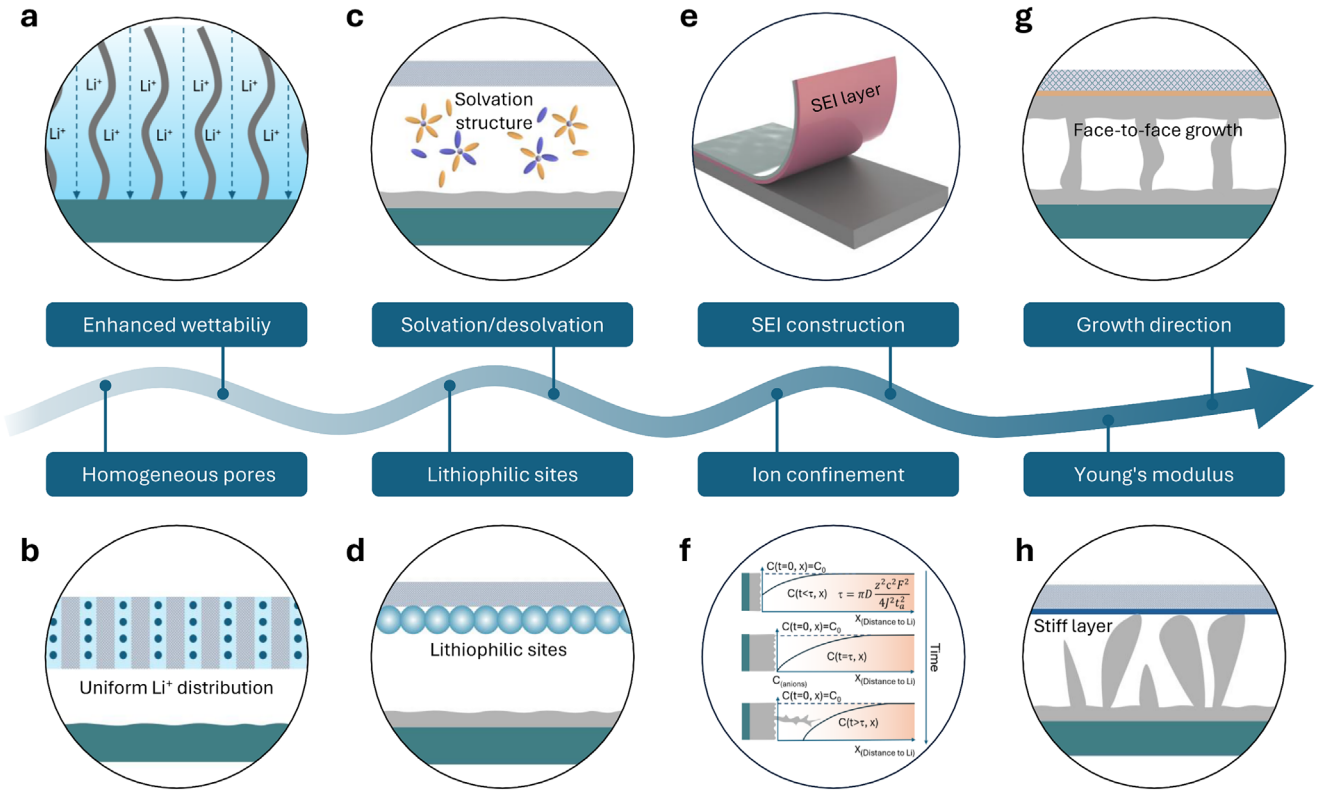


Figure 6. Schematic illustration of the separator functions for anodes in lithium-ion batteries (LIBs) by tailoring a) wettability, b) pore structures, c) solvation/desolvation behaviors, d) lithiophilic sites, e) solid electrolyte interphase (SEI) layer, f) ion confinement, g) growth direction, and h) Young's modulus.

precursor, interconnected mesoporous structures with an average mesopore size of 21 nm can be generated by selectively thermally decomposing the PLA phase at 280 °C.[130] PDA-modified polyetherimide track-etched membranes (PEITEM) have been developed. The PDA@PEITEM ion management membrane features vertically aligned, uniform negatively charged nanochannels, which can act as an ion distributor.[131]

In addition to the design of the pore structure of the separator itself, it is also possible to load functional materials with uniform pores on the anode side of the separator to achieve ion flow regulation. Using silica as an example, the increased tortuosity caused by silica particles affects the ion diffusion path, leading to concentration polarization and battery performance degradation. To address this issue, Zhao's group designed a separator modified with hollow mesoporous silica (HMS) particles.[132] The HMS particles have a diameter of approximately 500 nm, a shell thickness of about 100 nm, and an internal cavity of about 300 nm. These HMS particles can significantly shorten the Li^+ conduction pathways through their mesoporous channels and internal cavities, thereby enhancing the electrochemical kinetics of the battery.

4.2.3. Separator-Induced Ion Solvation/Desolvation

The solvation structure of Li^+ ions depends on the environment in which they exist, including salt concentration, solvent type, and solute composition.[133] It determines the mobility of Li^+ in the electrolyte and separator, and further influences the composition of the SEI on the electrode. Specifically, the solvation structure can be altered by the surface charge of the separator, which can increase the migration rate of Li^+ and change the SEI composition (Figure 6c). In recent years, using materials with ordered pore structures (e.g., metal-organic frameworks and molecular sieves) to regulate the solvation structure of metal ions in the electrolyte, thereby constructing an ideal SEI layer, has become one of the research hotspots in lithium metal batteries.

For example, a metal-organic cluster (Zr-MOC) was coated on the conventional separator to inhibit the reactivity between lithium metal and carbonate electrolytes.[134] The nanostructure of the separators can effectively modulate the energy state of Li^+ and induce the solvent molecules to partially remove from their primary solvation structure, thus reducing the difficulty of Li^+ reduction and making the reduction of Li^+ much easier. Similarly, a freestanding COF separator (TPB-BD$(OH)_2$-COF) was synthesized and used as a separator in lithium metal batteries.[135] The hydroxyl functional groups (-OH) in TPB-BD$(OH)_2$-COF form hydrogen bond networks with PF_6^-, EC, and EMC in the ester-based electrolyte, creating -OH···F and -OH···O interactions. This process facilitates the Li^+ solvation sheath to achieve desolvation and the formation of a locally concentrated electrolyte, which inhibits the side reaction of the free solvent with lithium metal. Apart from framework materials, lithium montmorillonite (LiMMT) was also utilized as a prototype to construct a functional coating on a commercial separator. As Lewis

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acid sites and electron acceptors, the abundant -OH groups in LiMMT selectively adsorb the corresponding FSI^- anions, thereby modulating the solvated shell structure and restricting their migration.[136]

4.2.4. Lithiophilic Sites

The key mechanism involves constructing lithiophilic functional materials that optimize lithium deposition sites, electrolyte wettability, Li^+ conductivity, and transference number, thereby achieving uniform lithium deposition (Figure 6d). Typical functional materials include lithiophilic carbon-based materials (e.g., carbon nanosheets with cubic cavity,[137] CNT,[138] and carbon dots[139]), phosphorus-based materials,[140] polymer-based materials (e.g., donor–acceptor polymer,[141] and poly(acrylonitrile) nanosheets[142]), metal compound-based materials (e.g., Fe_3N@N-doped graphene[143]), metal-organic frameworks (e.g., MOF801,[144] and UIO-66-F[145]) and covalent organic frameworks (e.g., F-BtTp-Nfilm,[146] PA-COF[147] and zwitterionic COF[148]). The design of these materials focuses on creating lithiophilic sites and Li^+ transport channels, which help regulate the Li^+ flux or concentration on the lithium metal anode surface. This effectively promotes a uniform and dense lithium deposition behavior and enhances the transport and migration kinetics of Li^+, fundamentally inhibiting dendrite formation and growth. By solving the issues associated with lithium metal, these materials promote the electrochemical performance of the batteries. The critical aspects of designing these functional materials include the construction of Li^+ transport channels and oriented anchoring sites.

4.2.5. Constructing Artificial SEI

An uneven and unstable SEI can trigger uncontrollable lithium dendrite growth.[133] On the one hand, the heterogeneous structure of the SEI exhibits poor mechanical properties, failing to meet the requirements for a stable SEI. This makes it susceptible to being punctured by lithium dendrites and detaching from the current collector, leading to the formation of a large amount of dead lithium and accelerating dendrite growth. On the other hand, poor ionic conductivity of an unstable SEI results in slow Li^+ transport kinetics, causing excessive lithium dendrite growth, increased interfacial impedance, and higher polarization voltage.[149] Therefore, using functionalized separators to assist in the formation of a stable SEI is an effective strategy to suppress lithium dendrite growth (Figure 6e). Research has shown that the key to forming a stable and highly ion-conductive SEI is to create components that promote a uniform structure and enhanced ionic conductivity, such as LiF, Li_3N, Li_5Sn_2 alloys, LiN_xO_y, and lithiophilic metals (e.g., Mg and Sn). For this strategy, designing and preparing new functional materials that promote the formation of stable SEI components to modify the separator is a critical and focal point.

4.2.6. Anion-Ion Fixing

The separator and its modified materials can interact with the electrolyte. These interactions greatly affect the behavior of lithium salt ions and electrolyte solvents, leading to changes in the Li^+ transference number (Figure 6f). The sand time is widely recognized as the time point when cation depletion near the plating electrode is completed, which is consistent with the time of dendrite nucleation.[150] Therefore, extending the sand time can reduce the formation of dendrites.[151] The expression for sand time is:

$$\tau = \pi D\left(\frac{C_0 e}{2Jt_a}\right)^2 \tag{3}$$

$$t_a = 1 - t_{Li^+} = \frac{\mu_a}{\mu_a + \mu_{Li^+}} \tag{4}$$

Among which, D is the bipolar coefficient, C_0 is the initial Li^+ concentration, e is the transfer charge, J is the current density, t_a and t_{Li^+} are the transference numbers of cations and anions, μ_a and μ_{Li^+} are the mobilities of cations and anions. By fixing the anions in the electrolyte to prolong the formation time of the space charge region, the onset of dendrite nucleation can be delayed. Currently, a significant amount of work is focused on designing cation-selective (CS) separators to enhance the Li^+ transference number. For example, researchers developed a large-area CS separator construction strategy based on a PVDF membrane to improve the performance of lithium metal anodes.[152] This CS separator not only significantly increases the Li^+ transference number ($t_{Li^+} = 0.81$) but also substantially reduces the anion transference number, thereby extending the nucleation time of Li dendrites. Similarly, using specific functional groups to restrict the movement of anions is also a common approach to address this issue.[153] These functional groups can interact with anions, either through electrostatic interactions or by forming coordination complexes, thereby reducing their mobility and enhancing the Li^+ transference number. This strategy helps to stabilize the electrolyte and delay the formation of lithium dendrites, contributing to the overall safety and performance of lithium metal batteries.

4.2.7. Controlling Growth Direction

In a conventional sense, dendrites grow from the surface of lithium metal toward the separator. However, by designing the separator side to be conductive, the growth direction of lithium metal can be changed (Figure 6g). Unlike conventional methods that attempt to inhibit or eliminate dendrite formation, this approach works to convert low-dimensional fibrous dendrites into high-dimensional lithium metal layers.

For example, functionalized nanocarbon (FNC) was coated on the separator to induce Li deposition.[154] After cycling, lithium dendrites are grown from the FNC layer on the separator and the lithium metal anode simultaneously in opposite directions. This method not only avoids the short circuits but also accommodates the volume expansion behavior of the SEI film during Li plating/stripping cycling without increasing the impedance of the nanocarbon layer. Apart from carbon materials, metals are usually recognized as potential alternatives. As a lithiophilic material, Au can attract Li^+ to deposit uniformly on one side of the separator rather than on the current collector side.[155] By coating a PP separator with a 13.4 nm layer of gold nanoparticles,

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uniform lithium metal deposition was achieved, and the lithium was successfully guided to grow in a direction opposite to the conventional one. The design not only effectively prevents lithium dendrites from penetrating the separator but also transforms the morphology of the deposited lithium from loose dendritic to dense and compact.

4.2.8. Realizing High Modulus

The mechanical strength of current commercial separators is insufficient to resist the penetration of lithium dendrites.[156] Lithium dendrites are prone to penetrate the separators during cycling and cause short circuits, which pose significant safety risks. Therefore, the development of a new type of separator with ultrahigh mechanical strength is required to delay the time required for dendrites to penetrate the separator, thereby extending the life of lithium metal batteries (Figure 6h). Researchers have shown that separators with a high shear modulus (over 7 GPa, 1.8 times that of lithium metal) can prevent dendrite propagation.[157] Currently, ultrahigh shear modulus polymer fibers, such as Kevlar,[158] polybenzimidazole (PBI),[9] and poly(p-phenylene benzobisoxazole) (PBO) nanofibers,[159] have been used to prepare dendrite-resistant separators. Additionally, if the local mechanical properties of the separator are sufficiently excellent, they can also inhibit the growth of lithium dendrites. Therefore, preparing polymer separators with high modulus or modifying the separator with inorganic particles can also effectively prevent dendrites from penetrating the separator.

Currently, ceramic nanoparticle coatings are widely used to enhance the thermal stability and electrolyte wettability of polyolefin separators. However, force analysis indicates that nanoparticle coatings struggle to effectively resist localized external impacts, which inevitably leads to nonuniform Li^+ flow within the battery during charging and discharging processes, causing uneven lithium deposition on the electrodes and even the formation of lithium dendrites. Based on a deep understanding of the high toughness principle of nacre in nature, the research team constructed a nacre-inspired "brick-and-mortar" ordered structure on the surface of a PE separator.[160] When subjected to external impact, the nacre-inspired coating effectively dissipates the impact stress by increasing the area over which the force is distributed through a sliding mechanism, thus protecting the internal pore structure of the separator and maintaining a uniform Li^+ flow within the battery.

5. High Thermal Safety Separator

When LIBs are in an abusive state, the internal temperature of the battery will rise and trigger a series of exothermic side reactions. The heat accumulated from these reactions will exacerbate the thermal runaway of the battery and cause the organic electrolyte to ignite, ultimately leading to fire and explosion. As the carrier of the electrolyte, the separator can significantly influence the battery's operating condition at high temperatures by regulating its physicochemical properties. In this section, we mainly analyzed the thermal effects of the separator in batteries, whose functions primarily include thermal stability, thermal conduction, thermal buffer, and thermal protection.

5.1. Thermal Stability

The thermal stability of the separator refers to its ability to maintain its physical and chemical properties under high-temperature conditions. This is crucial for preventing thermal runaway in batteries under high-temperature environments. The selection and design of separator materials must ensure that they remain stable within the normal operating temperature range of the battery and under extreme temperature conditions that may be encountered. Currently, the design and development of thermally stable separators focuses on two main strategies: integrating thermally stable materials and constructing high heat-resistant skeletons. These innovations are designed to radically improve separators' stability at high temperatures, opening up new paths for the safe application of battery technology.

Although the polyolefin separator has superior chemical stability (PE/130 °C, PP/165 °C), the separator shrinks significantly under extreme conditions, making it difficult to guarantee the safety of the battery. In order to enhance the thermal resistance of the polyolefin separator, ceramic particles such as Al_2O_3,[161] ZrO_2,[162] SiO_2,[11] and zeolite[163] are introduced on the surface of the polyolefin substrate as a thermally stable coating. This strategy not only enhances the wettability of the separator but also significantly improves the safety of the battery. However, it is worth noting that many ceramic-coating separators still suffer from a lack of thermal stability. This is mainly attributed to the fact that conventional inorganic-coating separators often use low-melting-point hydrophilic polymers as binders, such as PMMA, PVDF-HFP, and PVDF. When the binder reaches its melting point, the bond between the coating and separator may be reduced, even leading to aggregation of ceramic particles in the melted polymer binder, which reduces the heat resistance of the coating. Therefore, the use of high melting-point polymers to substitute conventional binders or in situ preparation methods are required to be developed.[164]

To avoid the risks mentioned above, constructing a high heat-resistant skeleton can significantly increase the heat deformation temperature of the separators, ensuring that they can still maintain structural stability under extreme conditions, thus effectively avoiding the risk of short circuits. For polymer separators, when the temperature is below the glass transition temperature (T_g), neither the molecular chains nor the chain segments are able to move, and the polymer exhibits a glassy state, with only the atoms (or groups) that make up the molecule vibrating around their equilibrium positions. As the temperature continues to rise above T_g, the entire molecular chain begins to move, showing a rubbery state, with the deformation gradually increasing and becoming irreversible, leading to a direct short circuit between the positive and negative electrodes (**Figure 7a**). Therefore, in order to improve the thermal stability of commercial polyolefin separators, researchers have developed new separators by introducing heat-resistant polymers, such as polyimide (PI),[165] polyether ether ketone (PEEK),[166] PBI,[167] PBO,[168] and polyarylene ether nitrile (PEN),[169] which have excellent flame retardancy and stability. These materials can be processed into separators through specific methods to enhance the thermal stability of the battery, which is crucial for preventing separator shrinkage and flame retardancy in high-temperature abuse conditions. Taking PI as an example, it has a glass transition temperature of around 260 °C

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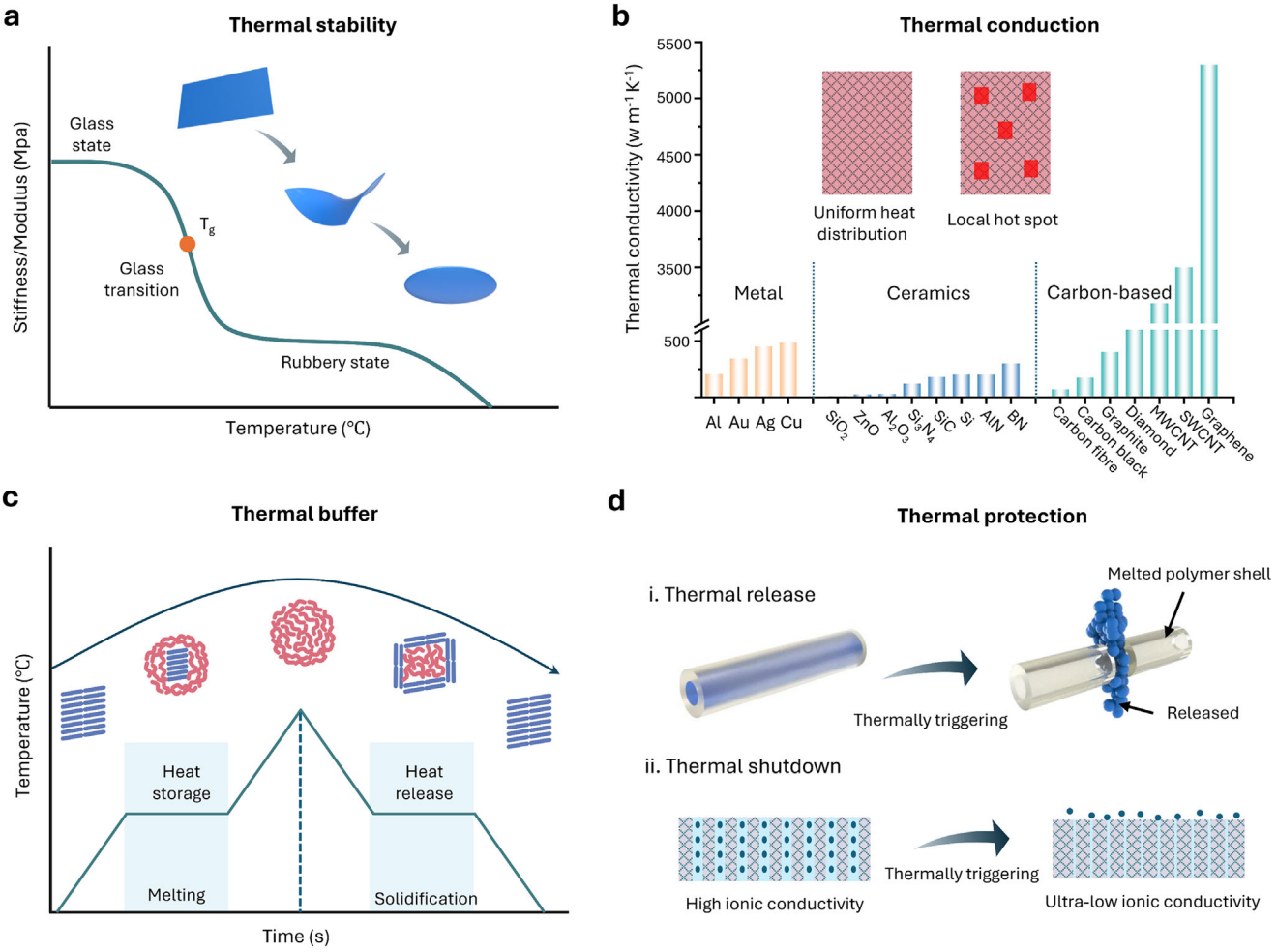


Figure 7. a) Changed polymer states with temperature, b) comparison of various materials on thermal conductivity, c) mechanism of the thermal buffer separator for alleviating the internal temperature, and d) schematic illustration of thermal release and thermal shutdown of separators.

and a thermal decomposition temperature greater than 500 °C. Yao's group prepared a PI separator that allows LIBs to retain 85.1% of their capacity after 200 cycles.[170] In contrast, batteries using traditional separators experience rapid capacity decay and micro-short circuits. Furthermore, soft-pack batteries using the PI separator can operate normally even at 150 °C.

5.2. Thermal Conductivity

Temperature variations can significantly affect the kinetics of electrochemical reactions and Li^+ transport. Studies have shown that a uniform temperature distribution inside the cell is a key factor in inhibiting the growth of lithium dendrites, ensuring safety under high temperatures.[171] However, commercial polyolefin separators are difficult to achieve ideal thermal management due to their low thermal conductivity (<1.0 W m^{-1} K^{-1}). For this reason, researchers have modified polyolefin separators by introducing thermally conductive materials to achieve uniform thermal diffusion to stabilize the redox reaction.

The thermal conductivity of typical materials including metal, ceramics, and carbon-based materials are compared in Figure 7b.[172] Among ceramics, BN,[173] AlN,[174] and SiC[175] have attracted much attention due to their superior thermal conductivity. Specifically, AlN is an ideal material for separator modification, which can effectively homogenize the temperature distribution of the battery and promote uniform nucleation of Li^+. When a local heat source is incident on the separator, the heat is able to diffuse quickly through the BN-modified separator, thereby reducing the temperature peak in the central region. For instance, Wu's group has developed an innovative method to combine PBI with AlN nanowires to prepare PBI-AlN composite membrane separators.[176] This method aims to improve the safety and performance of LIBs. By using infrared laser-generated hotspots to map the temperature distribution of different separators, the study shows that the PBI-AlN700 separator significantly reduces the central hotspot temperature to 60 °C, effectively improving the uniformity of the temperature distribution. This effect is notably better than that of pure PBI and PP separators, where the central hotspot temperatures are approxi-

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mately 80 °C and 90 °C, respectively, with larger temperature gradients.

In addition to insulating materials, conductive carbon materials also exhibit excellent thermal conductivity. This property makes them valuable for applications where both electrical conductivity and heat dissipation are important. Considering the extremely high thermal conductivity of graphene flakes at room temperature, ranging from 3080 to 5150 W m^{-1} K^{-1}, by constructing a thermally conductive coating of graphene, the authors successfully prepared a separator that eliminates localized hot spots.[177] In this design, the graphene side faces the lithium metal, serving as an in situ heat dissipation medium. Dissipation of localized heat accumulation around irregular lithium deposition inhibits further growth of lithium dendrites. While thermally conductive separators help to achieve a more uniform thermal distribution and eliminate heat accumulation, thereby reducing the number of cracks in the SEI film and inhibiting dendrite growth, they only promote uniform heat distribution and reduce localized heat. However, if the total heat generated within the battery cannot be dissipated in a timely manner, the overall temperature may rise rapidly, leading to safety issues.

5.3. Thermal Buffer

Phase change materials (PCMs) are reversible energy storage materials that can periodically store energy in the form of latent heat during phase transitions, achieving energy savings in practical applications (Figure 7c).[178] Organic-based phase change materials (OPCMs) with phase change temperatures in the mid-to-low range possess long-term chemical stability, nontoxicity, noncorrosiveness, environmentally friendliness, and efficiency. These materials are primarily categorized based on their chemical composition into paraffin (e.g., alkanes and their mixtures) and nonparaffin (e.g., fatty acids, alcohols, and glycols) types. Currently, alkanes, fatty acids, and polyols are the three major organic-based PCMs used for mid-to-low temperature thermal energy storage.

Paraffin, as a thermal energy storage PCM, has been widely studied for use in external battery thermal management systems (BTMSs).[179] Despite its advantages, paraffin presents significant technical challenges: its inherently low thermal conductivity (typically 0.1–0.3 W m^{-1} K^{-1}) coupled with high thermal resistance, tendency for liquid leakage during melting, and poor suitability for thin-film processing. Integrating paraffin into the complex electrochemical environment of batteries presents a significant challenge. Additionally, the internal space of a battery is limited, so effectively increasing the spatial utilization of paraffin to enhance thermal regulation capability is also crucial.

Encapsulating paraffin directly within the separator polymer is an effective approach. For example, using the co-axial electrospinning technology, paraffin PCM was encapsulated in hollow PAN nanofibers to prepare PW@PAN phase change nanofibers.[180] Under abusive conditions, the heat generated within the battery stimulates the encapsulated PCM to melt, thereby absorbing a large amount of heat without causing a significant temperature increase. Nail penetration tests simulating internal short circuits in prototype pouch cells demonstrated that the PCM-based

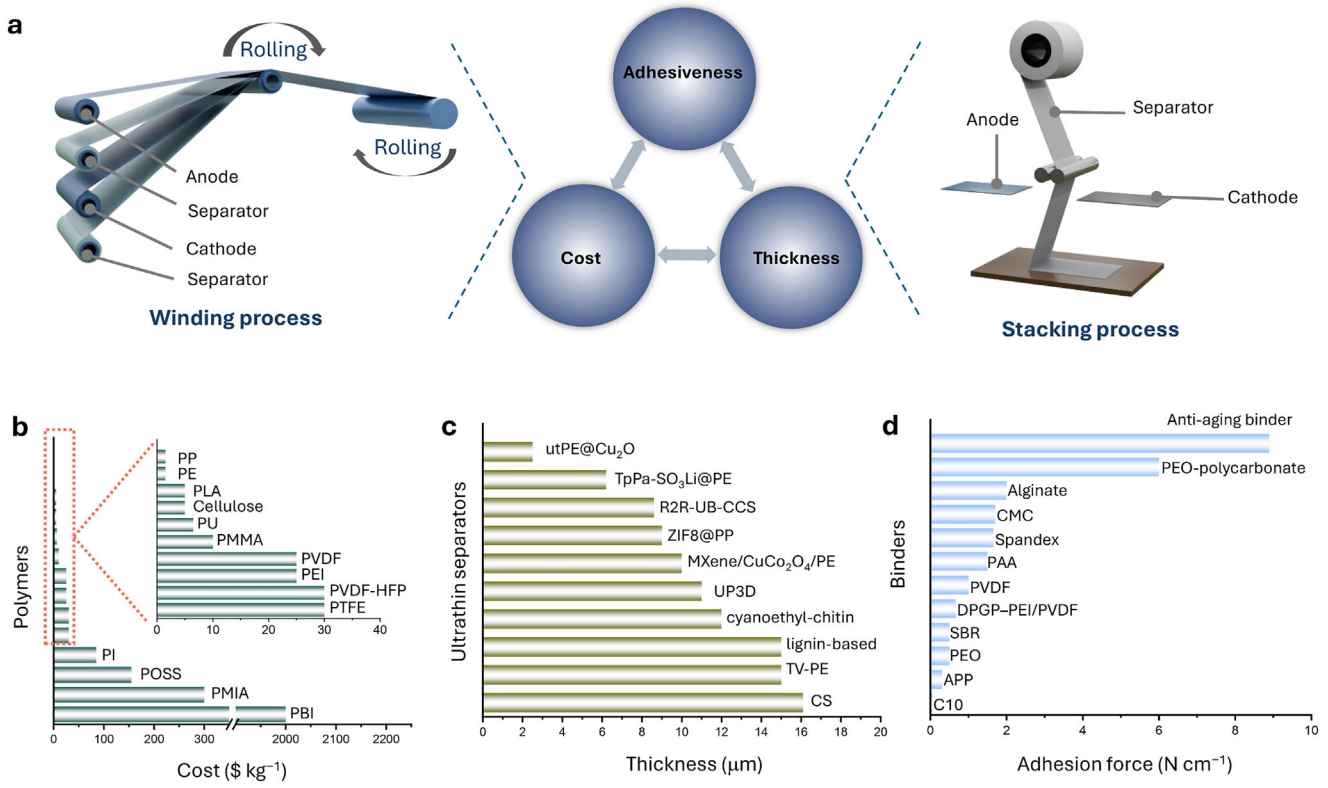


Figure 8. a) The preparation process of commercial lithium-ion batteries (LIBs), b) comparison of polymer costs, c) up-to-date work with a separator thickness of less than 20 μm, and d) comparison of the adhesion forces of various binders.

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separator can effectively suppress temperature increases caused by battery failures. Additionally, owing to the unique heat storage capability of the PCM separator, the penetrated battery can rapidly cool down to room temperature within 35 seconds.

In addition to paraffin, stearic acid (SA) has a melting point of 72 °C, which is close to the initial temperature at which batteries begin to experience abnormal heat generation. This makes SA particularly suitable for use in battery thermal management systems, as it can effectively absorb and dissipate heat when the battery temperature starts to rise, thereby preventing further temperature increases and potential thermal runaway. For example, encapsulating triethyl phosphate and SA into hollow porous silica microspheres and coating them on the separator surface can prevent them from dissolving directly into the electrolyte, ensuring that the battery's electrochemical performance is not affected.[181] Under thermal stimulation, the PCM undergoes a solid-liquid phase transition, which promotes molecular migration and the release of flame retardants. The low melting point of PCM ensures that the flame retardants is released in time during the initial stage of thermal runaway in the battery, preventing internal short circuits caused by thermal inertia.

5.4. Thermal Protection

Thermal protection in batteries mainly consists of two forms: thermal release and thermal shutdown (Figure 7d). Once the metal dendrite penetrates the separator or the temperature rises sharply, the battery should be protected. Otherwise, direct contact with the electrodes will lead to thermal runaway, which may trigger an explosion in severe cases. Currently, more research focuses on adding flame retardants into the electrolyte.[182] However, this strategy will inevitably affect the viscosity and ionic conductivity of the electrolyte, thereby reducing the electrochemical performance of the battery. Therefore, integrating flame retardants into a separator is an alternative approach for thermal release.

Cui's group used the electrospinning technology to prepare a "core–shell" structured microfiber, where the flame retardant triphenyl phosphate (TPP) is located in the core of the fiber, which is encapsulated by a polymer shell of polyvinylidene fluoride-hexafluoropropylene (PVDF-HFP).[183] These fibers are randomly stacked during the electrospinning process, resulting in a self-supporting porous membrane. The gaps between the randomly stacked fibers provide channels for the free movement of Li^+. Under normal battery operation, the flame retardant is protected within the PVDF-HFP shell, which prevents direct contact with the electrolyte, thereby minimizing its negative impact on electrochemical performance. Once the battery experiences thermal runaway and the temperature rises, the PVDF-HFP shell partially melts, releasing the encapsulated TPP into the electrolyte to suppress combustion.

In addition to phosphorus-containing compounds, bromine-based additive flame retardants such as decabromodiphenylethane (DBDPE) and decabromodiphenyl oxide (DBDPO) have been used to suppress thermal runaway in batteries. However, these materials can only inhibit chemical chain reactions in the gas phase and not in the electrolyte (i.e., the condensed phase). A dual-mechanism self-extinguishing separator composed of DBDPE and calcium oxide (CaO) has been designed.[184] Metal halide particles can neutralize hydrogen radicals in the condensed phase and remain solid at high temperatures, addressing the shortcomings of DBDPE and DBDPO in flame retardancy properties. The high bromine content (>81.5 wt%) of DBDPE can provide bromine under abnormal conditions. The high reactivity of CaO with bromine easily forms, $CaBr_2$, which can extinguish flames within 0.5 s. The DBDPE-CaO coated separator can generate gaseous hydrogen bromide (HBr) and solid $CaBr_2$, effectively inhibiting chain reactions in both the gas and liquid phases, thus exhibiting excellent flame-retardant properties.

As for the thermal shutdown, the mechanism involves the addition of low melting point materials (e.g., PE, PP, PMMA, PEI, and ethylene vinyl acetate copolymer) to the separators when the temperature of the cell rises to a critical value, the materials melt and form an insulating barrier to close the ion channel of the separator, interrupting the battery reactions. Shi et al. used PP fibers as a low melting point material and PAN fibers as a higher melting point material to act as a backbone.[185] After thermal treatment at 160 °C, the PP fiber material melted and diffused into the PAN fibers, blocking the pores of the separators and decreasing the electrical conductivity. After thermal treatment for 1 h, the ionic conductivity of the fiber membrane decreased from 1.99 to 0.32 mS cm^{-1} to achieve the thermal shutdown function.

6. Conclusion, Challenges, and Outlook

6.1. Conclusion

In LIBs, the separator exhibits dual functions as the physical barrier between the electrodes and a guarantee of efficient ion transport. Various properties of the separator, such as thickness, porosity, wettability, mechanical properties, and thermal dimensional stability, significantly affect the overall battery performance. However, traditional separators are no longer sufficient to meet the performance requirements of more advanced LIBs, and the improvement of existing separators and the development of new alternatives have become urgent tasks in this field. Researchers have made extensive efforts to optimize conventional separators, and in this review, we first summarized the basic characteristics and physicochemical properties of the separator, which form the foundation for guiding separator design. Subsequently, we concluded the preparation processes and techniques of the separator and conducted a comparative analysis of their advantages and drawbacks. Then, the roles of the separator in the cathode and anode of LIBs were emphasized and reviewed, accompanied by the latest research progress. Considering the critical role of the separator in battery safety, we also categorized and analyzed its functions in safety, particularly in thermal management. This work provides valuable references and guidance for the design and functional development of separators.

6.2. Challenges and Outlook

In this section, we highlight key issues that are rarely addressed in separator research, as well as potential research directions and factors that need to be considered in practical applications.

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i) Remaining challenges in separator standardization: Functional separators face challenges in commercialization despite various designs. The current battery assembly mainly uses a winding and stacking process (**Figure** 8a). Achieving commercial viability requires balancing cost, thickness, and adhesion while maintaining functionality. First, cost is a critical parameter for reducing battery costs and achieving large-scale production. As shown in Figure 8b, PP/PE remains dominant and serves as the baseline separator for further modification due to low cost. These newly designed polymers are expensive and the corresponding new manufacturing processes (e.g., electrospinning) are costly as well, far surpassing traditional dry and wet methods. Functional coatings, though promising, introduce thickness-related challenges, potentially impairing electrolyte absorption and ion transport. Moreover, excessive or uneven distribution of coatings can block pores, resulting in difficulties in electrolyte penetration, which in turn reduces the battery performance. Precise control of coating thickness and pore structure is essential to maintain battery performance. Recent developments in ultrathin separators (Figure 8c) include MXene/$CuCo_2O_4$/PE,[186] utPE@Cu_2O,[187] ZIF8@PP,[188] lignin-based ultrathin separator,[189] UP3D,[190] TpPa-SO_3Li@PE,[191] CS,[152] R2R-UB-CCS,[192] TV-PE,[193] and cyanoethyl-chitin nanofiber.[194] In the future, the thickness is expected to be reduced to less than 5 µm. Another important issue is the delamination of coatings from the separator substrate. In many cases, especially under thermal or mechanical stress, coatings may easily delaminate or peel off the separator substrate. It necessitates improvements in adhesion to ensure that they remain reliably fixed to the separator throughout the battery's lifecycle. Figure 8d compares the adhesive force of commonly used binders.[195] The research of new adhesion promoters, the exploration of advanced coating technologies and the development of multifunctional materials can help to solve these problems.

ii) The prerequisite for scalable production of functionalized separator: Many advanced separator designs rely on complex architectures, including multilayer coatings, nanoparticle

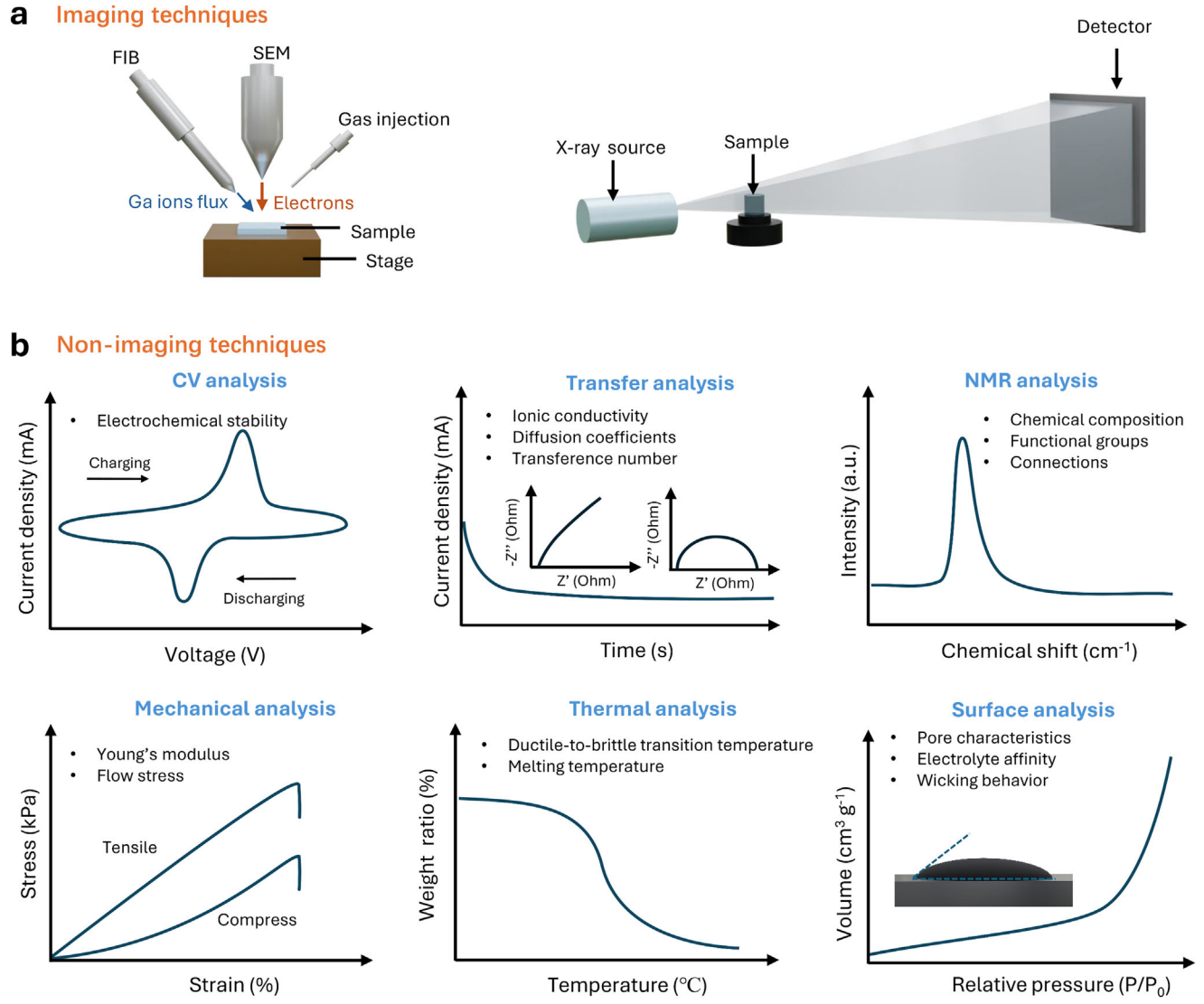


Figure 9. a) Imaging and b) non-imaging techniques for separator characterization.

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composites, or electrospun nanofiber matrices, which often involve intricate processing steps. These are generally difficult to replicate reliably in high-throughput production. Furthermore, the integration of additional functionalities, such as flame-retardant additives or polysulfide-blocking layers, can introduce new variables that affect processability, requiring careful optimization of parameters such as drying kinetics, slurry viscosity, and roll-to-roll handling. The mass production of functionalized battery separators necessitates a holistic approach that harmonizes material innovation, advanced manufacturing, and stringent performance validation.

iii) Post-mortem analysis for separator failure mechanisms in high-energy batteries: In high-energy batteries, separator failure predominantly arises from an interdependent degradation triad of pore collapse, electrolyte depletion, and thermal shrinkage, creating a self-amplifying failure cascade. Continuous electrolyte consumption through SEI growth and gas evolution generates localized dry zones that induce lithium plating while simultaneously triggering polymer matrix softening and pore collapse under thermal stress, severely constricting ion transport pathways. This structural degradation is exacerbated by separator shrinkage during cycling, which not only reduces interfacial contact but also increases current heterogeneity, further accelerating electrolyte depletion through intensified side reactions. The resulting accumulation of decomposition products progressively blocks remaining pores, creating a feedback loop where each collapsed pore increases local impedance, generates more heat, and worsens shrinkage, ultimately culminating in complete pore structure collapse and catastrophic dendrite penetration. These coupled mechanisms are particularly acute in high-voltage systems where thinner electrolytes and expanded thermal cycling amplify the rate of pore degradation, making the separator's structural integrity the critical bottleneck in battery failure.

Future separator designs must combine dendrite-resistant coatings, electrolyte-regulating architectures (e.g., hierarchical pores, absorbent nanoparticles), and self-stabilizing interfaces to disrupt failure feedback loops. A systems approach is essential, as isolated solutions cannot address the coupled degradation mechanisms in high-energy batteries. In situ and operando techniques, such as focused ion beam-scanning electron microscopy (FIB-SEM) and X-ray tomography (**Figure 9a**), are critical for visualizing Li deposition and separator degradation across scales.[196] Complementary nonimaging techniques such as cyclic voltammetry (CV), electrochemical transport, nuclear magnetic resonance (NMR), and mechanical/thermal testing (Figure 9b) quantify transport properties and interfacial stability. Integrating these datasets with accurate multiscale computational models is vital to predict localized aging and deformation, enabling rational separator optimization.

iv) Precision separator engineering for fast-charging and anode-free batteries: First, for fast-charging batteries, separators must ameliorate the potential lithium plating due to the huge hysteresis at >3C rates. The superior pore uniformity and asymmetric coatings on the anode side would be helpful to accelerate the Li^+ ion transport and to physically block the formed dendrites from short-circuiting. Additionally, electrolyte wettability must be enhanced, possibly through plasma-treated 3D pore architectures to prevent Li^+ ion depletion at high currents. Moreover, the fast-charging process would be accompanied by increased localized temperature, necessitating good thermal management. For instance, introducing thermally conductive fillers could dissipate the heat generation at high rates. For the specific separator

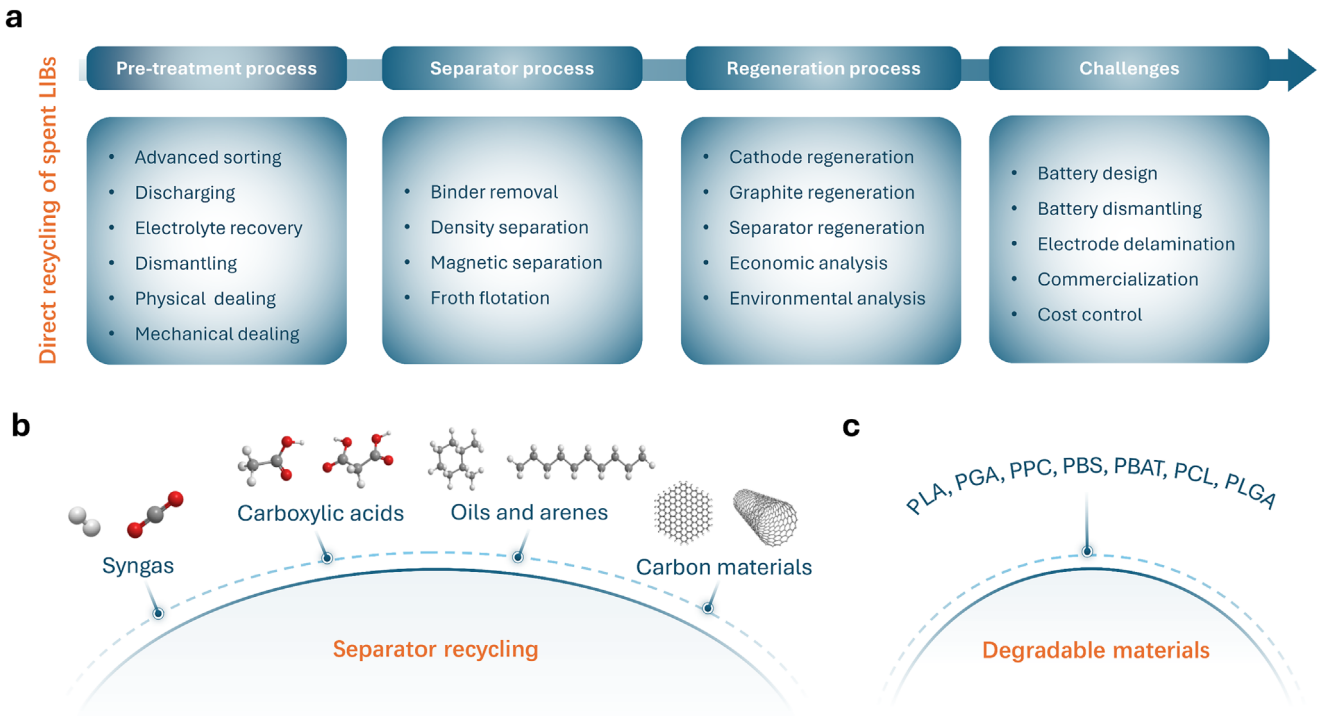


Figure 10. a) Direct recycling process for spent lithium-ion batteries (LIBs), b) possible products obtained from separator recycling, and c) degradable polymers.

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design in anode-free configurations, the zero-lithium-start conditions should be borne in mind, which requires a lithiophilic layer on the anode side to guide uniform Li deposition. Furthermore, it is better to consider an elastic buffering zone for stress dissipation to accommodate possible uneven Li growth. Finally, to enhance the energy density of anode-free batteries, optimal electrolyte retention is crucial. Innovative approaches, such as incorporating ionic liquid reservoirs, could effectively sustain continuous ionic contact throughout the repeated stripping and plating processes.

v) Sustainable separator recycling and treatment: Battery recycling has become a hot research topic, and considerable efforts have been devoted to developing recycling technologies for cathodes and anodes of batteries (**Figure 10a**).[197] However, there is less reporting on the recycling of separators. For separators, major polymer materials such as PP and PE can be recycled through pyrolysis, chemical, and high-temperature treatments for use in producing new products, including syngas, carboxylic acids, oils, and arenes, and carbon materials (Figure 10b).[198] To enhance sustainability, green chemistry approaches should be employed in separator recycling, including minimizing hazardous solvent usage or developing solvent-free processing technologies. Pollution control measures should also be strengthened, such as installing efficient exhaust gas treatment equipment. In addition, in order to directly and effectively reduce the pollution problem of separators to the environment, directly using degradable materials for separator design is also a promising research direction (Figure 10c).

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

anodes, cathodes, high thermal safety, lithium-ion batteries, separator

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[1] a) R. Wang, Y. Zhu, S. Huang, J. Fu, Y. Zhou, M. Li, L. Meng, X. Zhang, J. Liang, Z. Ran, M. Yang, J. Li, X. Dong, J. Hu, J. He, Q. Li, *Nat. Mater.* **2025**, *24*, 1074; b) P. Poizot, J. Gaubicher, S. Renault, L. Dubois, Y. Liang, Y. Yao, *Chem. Rev.* **2020**, *123*, 6490; c) M. Tang, Q. Liu, X. Zou, B. Zhang, L. An, *Adv. Mater.* **2025**, 2501361.

[2] a) D. M. Davies, M. G. Verde, O. Mnyshenko, Y. Chen, R. Rajeev, Y. Meng, G. Elliott, *Nat. Energy* **2018**, *4*, 42; b) S. W. D. Gourley, R. Brown, B. D. Adams, D. Higgins, *Joule* **2023**, *7*, 1415; c) Q. Liu, Z. Yu, Q. Zhuang, J. K. Kim, F. Kang, B. Zhang, *Adv. Mater.* **2023**, *35*, 2300498.

[3] a) Q. Wang, C. Zhao, J. Wang, Z. Yao, S. Wang, S. G. H. Kumar, S. Ganapathy, S. Eustace, X. Bai, B. Li, M. Wagemaker, *Nat. Commun.* **2023**, *14*, 440; b) J. Min, S. Bak, Y. Zhang, M. Yuan, N. F. Pietra, J. A. Russell, Z. Deng, D. Xia, L. Tao, Y. Du, H. Xiong, L. Li, L. A. Madsen, F. Lin, *Nat. Nanotechnol.* **2025**, *20*, 787; c) Q. Liu, Z. Zheng, P. Xiong, C. Huang, S. Huang, B. Zhao, Y. Wu, Y. Zhang, B-K. Kim, X. Yu, H. Park, *Electron* **2024**, *2*, 72.

[4] a) J. Hu, H. Wang, B. Xiao, P. Liu, T. Huang, Y. Li, X. Ren, Q. Zhang, J. Liu, X. Ouyang, X. Sun, *Natl. Sci. Rev.* **2023**, *10*, nwad252; b) H. Raza, S. Bai, J. Cheng, S. Majumder, H. Zhu, Q. Liu, G. Zheng, X. Li, G. Chen, *Electrochem. Energy Rev.* **2023**, *6*, 29; c) B. Song, L. Su, X. Liu, W. Gao, T. Wang, Y. Ma, Y. Zhong, X.-B. Cheng, Z. Zhu, J. He, Y. Wu, *Electron* **2023**, *1*, 13.

[5] a) S. Xia, X. Wu, Z. Zhang, Y. Cui, W. Liu, *Chem* **2019**, *5*, 753; b) L. Hou, X. Zhang, B. Li, Q. Zhang, *Mater. Today* **2021**, *45*, 62.

[6] a) Y. Feng, L. Zhou, H. Ma, Z. Wu, Q. Zhao, H. Li, K. Zhang, J. Chen, *Energy Environ. Sci.* **2022**, *15*, 1711; b) A. K. Thakur, A. Kumar, H. Park, H. Kim, M. S. Ahmed, A. M. Saleque, M. P. Vikram, R. Saidur, Y. Ma, J. Hwang, *J. Energy Storage* **2023**, *73*, 108873.

[7] a) D. Chen, Y. Liu, C. Feng, Y. He, S. Zhou, B. Yuan, Y. Dong, H. Xie, G. Zneg, J. Han, W. He, *Electron* **2023**, *1*, 1; b) Y. Ji, L. Dong, J. Liu, H. Xie, S. Zhong, C. Yang, J. Han, W. He, A. Li, *Energy Environ. Sci.* **2024**, *17*, 4078.

[8] a) M. F. Lagadec, R. Zahn, V. Wood, *Nat. Energy* **2018**, *4*, 16; b) Y. Zong, H. He, Y. Wang, M. Wu, X. Ren, Z. Bai, N. Wang, X. Ning, S. X. Dou, *Adv. Energy Mater.* **2023**, *13*, 2300403.

[9] Y. Deng, A. Hussain, W. Raza, X. Cai, D. Liu, J. Shen, *J. Energy Chem.* **2024**, *91*, 579.

[10] J. Gou, W. Liu, A. Tang, L. Wu, *J. Membrane Sci.* **2022**, *659*, 120807.

[11] Z. Fan, X. Chen, J. Shi, H. Nie, X. Zhang, X. Zhou, X. Xie, Z. Xue, *Nano-Micro Lett.* **2025**, *17*, 128.

[12] N. Lingappan, W. Lee, S. Passerini, M. Pecht, *Renewable Sustainbale Energy Rev.* **2023**, *187*, 113726.

[13] J. Li, H. Jia, S. Ma, L. Xie, X. Wei, L. Dai, H. Wang, F. Su, C. Chen, *ACS Energy Lett.* **2022**, *8*, 56.

[14] M. F. Lagadec, M. Ebner, R. Zahn, V. Wood, *J. Electrochem. Soc.* **2016**, *163*, A992.

[15] H. Lee, M. Yanilmaz, O. Toprakci, K. Fu, X. Zhang, *Energy Environ. Sci.* **2014**, *7*, 3857.

[16] a) K. Adámek, A. Havelka, Z. Kůs, A. Mazari, *Polymers* **2021**, *14*, 140; b) D. Djian, F. Alloin, S. Martinet, H. Lignier, J. Y. Sanchez, *J. Power Sources* **2007**, *172*, 416.

[17] M. Eftekharnia, M. Hasanpoor, M. Forsyth, R. Kerr, P. C. Howlett, *ACS Appl. Energy Mater.* **2019**, *2*, 6655.

[18] S. Huo, L. Sheng, W. Xue, L. Wang, H. Xu, H. Zhang, X. He, *InfoMat* **2023**, *5*, 12394.

[19] C. F. J. Francis, I. L. Kyratzis, A. S. Best, *Adv. Mater.* **2020**, *32*, 1904205.

[20] J. Xing, J. Li, W. Fan, T. Zhao, X. Chen, H. Li, Y. Cui, Z. Wei, Y. Zhao, *Composites, Part B* **2022**, *243*, 110105.

[21] K. Wang, J. Duan, X. Chen, J. Wang, J. Li, L. Jiang, W. Yan, W. Lyu, Y. Liao, *Adv. Energy Mater.* **2024**, *14*, 2401146.

[22] M. Waqas, S. Ali, C. Feng, D. Chen, J. Han, W. He, *Small* **2019**, *15*, 1901689.

[23] Y. Wu, F. Yang, Y. Cao, M. Xiang, J. Kang, T. Wu, Q. Fu, *Polymer* **2021**, *230*, 124081.

[24] X. Sun, Y. Zhou, D. Li, K. Zhao, L. Wang, P. Tan, H. Dong, Y. Wang, J. Liang, *Carbon Energy* **2024**, *6*, 539.

[25] Y. Li, Q. Li, Z. Tan, *J. Power Sources* **2019**, *443*, 227262.

[26] H. Xiang, F. Zhang, B. Zou, Q. Hou, C. Cheng, M. Lu, X. Wang, W. Ping, Y. Sun, X. Song, *J. Membrane Sci.* **2024**, *704*, 122867.

[27] Y. Liang, S. Cheng, J. Zhao, C. Zhang, S. Sun, N. Zhou, Y. Qiu, X. Zhang, *J. Power Sources* **2013**, *240*, 204.

[28] W. Tang, Q. Liu, N. Luo, F. Chen, Q. Fu, *Compos. Sci. Technol.* **2022**, *225*, 109479.

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[29] J. Xue, T. Wu, Y. Dai, Y. Xia, *Chem. Rev.* **2019**, *119*, 5298.
[30] B. Yang, Y. Yang, X. Xu, Y. Ke, Y. Pan, L. Su, Y. Wang, S. Wang, J. Qian, R. Xia, E. Fu, *J. Mater. Sci.* **2022**, *57*, 11274.
[31] M. Shin, W. J. Song, H. B. Son, S. Yoo, S. Kim, G. Song, N. S. Choi, S. Park, *Adv. Energy Mater.* **2018**, *8*, 1801025.
[32] A. Venault, C. Chang, T. Tsai, H. Chang, D. Bouyer, K. Lee, Y. Chang, *J. Membr. Sci.* **2018**, *563*, 54.
[33] D. M. Correia, C. Ribeiro, V. Sencadas, L. Vikingsson, M. O. Gasch, J. L. Gómez Ribelles, G. Botelho, S. Lanceros-Méndez, *Mater. Des.* **2016**, *92*, 674.
[34] T. Zhao, P. Xiao, S. Nie, M. Luo, M. Zou, Y. Chen, *Coord. Chem. Rev.* **2024**, *502*, 215592.
[35] a) X. Zhang, S. Cheng, C. Fu, G. Yin, L. Wang, Y. Wu, H. Huo, *Nano-Micro Lett.* **2024**, *17*, 2; b) A. Manthiram, X. Yu, S. Wang, *Nat. Rev. Mater.* **2017**, *2*, 16103.
[36] D. Cao, Y. Zhao, X. Sun, A. Natan, Y. Wang, P. Xiang, W. Wang, H. Zhu, *ACS Energy Lett.* **2020**, *5*, 3468.
[37] Z. Wu, G. Zeng, J. Yin, C. Chiang, Q. Zhang, B. Zhang, J. Chen, Y. Yan, Y. Tang, H. Zhang, S. Zhou, Q. Wang, X. Kuai, Y. Lin, L. Gu, Y. Qiao, S. Sun, *ACS Energy Lett.* **2023**, *8*, 4806.
[38] Y. Sun, Z. Chen, H. Noh, D. Lee, H. Jung, Y. Ren, S. Wang, C. S. Yoon, S. Myung, K. Amine, *Nat. Mater.* **2012**, *11*, 942.
[39] B. L. D. Rinkel, J. P. Vivek, N. Garcia-Araez, C. P. Grey, *Energy Environ. Sci.* **2022**, *15*, 3416.
[40] Y. Liu, C. Zeng, M. Hu, H. Liang, X. Wei, X. Ji, S. Wang, X. Liu, Y. Cui, J. Lin, T. Zhai, H. Li, *Adv. Energy Mater.* **2025**, *15*, 2405960.
[41] S. Zhang, N. S. Andreas, R. Li, N. Zhang, C. Sun, D. Lu, T. Gao, L. Chen, X. Fan, *Energy Storage Mater.* **2022**, *48*, 44.
[42] Y. Qiao, H. Yang, Z. Chang, H. Deng, X. Li, H. Zhou, *Nat. Energy* **2021**, *6*, 653.
[43] Y. Qiao, K. Jiang, H. Deng, H. Zhou, *Nat. Catal.* **2019**, *2*, 1035.
[44] Y. Zhan, H. Yu, L. Ben, Y. Chen, X. Huang, *Electrochim. Acta* **2017**, *255*, 212.
[45] C. Zhan, Z. Yao, J. Lu, L. Ma, V. A. Maroni, L. Li, E. Lee, E. E. Alp, T. Wu, J. Wen, Y. Ren, C. Johnson, M. M. Thackeray, M. K. Y. Chan, C. Wolverton, K. Amine, *Nat. Energy* **2017**, *2*, 963.
[46] M. S. Park, Y. G. Lim, J. H. Kim, Y. J. Kim, J. Cho, J. S. Kim, *Adv. Energy Mater.* **2011**, *1*, 1002.
[47] Y. Guo, X. Li, Z. Wang, H. Guo, J. Wang, *J. Energy Chem.* **2020**, *47*, 38.
[48] W. Zhong, R. He, L. Peng, W. Liu, J. Peng, H. Zhu, J. Xiang, S. Cheng, J. Xie, *Adv. Energy Mater.* **2025**, *15*, 2406007.
[49] Y. Mao, C. Zhou, H. Gong, S. Zhang, X. Wang, X. Liu, Q. Xiang, J. Sun, *Small* **2023**, *19*, 2303259.
[50] G. Liu, W. Wan, Q. Nie, C. Zhang, X. Chen, W. Lin, X. Wei, Y. Huang, J. Li, C. Wang, *Energy Environ. Sci.* **2024**, *17*, 1163.
[51] Q. Meng, M. Fan, X. Chang, H. Li, W. Wang, Y. Zhu, J. Wan, Y. Zhao, F. Wang, R. Wen, S. Xin, Y. Guo, *Adv. Energy Mater.* **2023**, *13*, 2300507.
[52] X. Chang, M. Fan, B. Yuan, W. He, C. Gu, C. Li, Q. Meng, Y. Guo, *Angew. Chem., Int. Ed.* **2024**, *63*, 202406557.
[53] J. Nguyen, Y. Lee, Y. Yang, *Small* **2024**, 2410453.
[54] R. Sim, L. Su, A. Dolocan, A. Manthiram, *Adv. Mater.* **2024**, *36*, 2311573.
[55] L. Rynearson, C. Antolini, C. Jayawardana, M. Yeddala, D. Hayes, B. L. Lucht, *Angew. Chem., Int. Ed.* **2024**, *63*, 202317109.
[56] Y. Feng, B. Zhong, R. Zhang, J. Yu, Z. Huang, Y. Huang, Z. Wu, Y. Fan, J. Tian, W. Xie, K. Zhang, *Adv. Energy Mater.* **2023**, *13*, 2302295.
[57] C. Han, Y. Cao, M. Yang, Y. Wang, D. Tang, S. Zhang, Y. Jia, Y. Zhang, H. Kim, F. Pan, Z. Jiang, J. Sun, *J. Energy Chem.* **2024**, *96*, 72.
[58] Q. An, Q. Liu, P. Mao, L. Duan, H. Zhu, L. Liu, G. Zhao, Y. Zha, L. Yang, M. Sun, Y. Fan, F. Xie, G. Hu, H. Guo, *Angew. Chem., Int. Ed.* **2025**, *64*, 202422539.
[59] a) J. G. Han, K. Kim, Y. Lee, N. S. Choi, *Adv. Mater.* **2019**, *31*, 1804822; b) K. Kim, D. Hwang, S. Kim, S. O. Park, H. Cha, Y. S. Lee, J. Cho, S. K. Kwak, N. S. Choi, *Adv. Energy Mater.* **2020**, *10*, 2000012.
[60] L. Ding, Y. Chen, Y. Sheng, X. Yue, Z. Liang, *Angew. Chem., Int. Ed.* **2024**, *63*, 202411933.
[61] M. Zhang, K. Liu, Y. Gan, H. Wang, F. Liu, M. Bai, X. Tang, Z. Wang, S. Li, A. Shao, K. Zhou, T. Wang, Z. Wang, S. Yuan, Y. Ma, *Adv. Energy Mater.* **2022**, *12*, 2201390.
[62] L. Sheng, K. Yang, J. Chen, D. Zhu, L. Wang, J. Wang, Y. Tang, H. Xu, X. He, *Adv. Mater.* **2023**, *35*, 2212292.
[63] S. K. Sandstrom, X. Ji, *Joule* **2023**, *7*, 13.
[64] X. Zhang, W. Guo, Y. Fu, *Acc. Mater. Res.* **2024**, *5*, 316.
[65] Z. W. Seh, Y. Sun, Q. Zhang, Y. Cui, *Chem. Soc. Rev.* **2016**, *45*, 5605.
[66] W. Tang, Z. Chen, B. Tian, H. Lee, X. Zhao, X. Fan, Y. Fan, K. Leng, C. Peng, M. Kim, M. Li, M. Lin, J. Su, J. Chen, H. Y. Jeong, X. Yin, Q. Zhang, W. Zhou, K. P. Loh, G. Zheng, *J. Am. Chem. Soc.* **2017**, *139*, 10133.
[67] Y. Matsuda, H. Nakashima, M. Morita, Y. Takasu, *J. Electrochem. Soc.* **1981**, *128*, 2552.
[68] Y. Li, W. Wang, X. Liu, E. Mao, M. Wang, G. Li, L. Fu, Z. Li, A. Y. S. Eng, Z. W. Seh, Y. Sun, *Energy Storage Mater.* **2019**, *23*, 261.
[69] J. Oh, H. Jo, H. Lee, H. T. Kim, Y. M. Lee, M. H. Ryou, *J. Power Sources* **2019**, *430*, 130.
[70] X. Ou, Y. Yu, R. Wu, A. Tyagi, M. Zhuang, Y. Ding, I. H. Abidi, H. Wu, F. Wang, Z. Luo, *ACS Appl. Mater. Interfaces* **2018**, *10*, 5534.
[71] X. Yu, H. Wu, J. H. Koo, A. Manthiram, *Adv. Energy Mater.* **2019**, *10*, 1902872.
[72] J. Yang, S. Zhao, Y. Lu, X. Zeng, W. Lv, G. Cao, *Nano Energy* **2020**, *68*, 104356.
[73] Z. Liang, C. Peng, J. Shen, J. Yuan, Y. Yang, D. Xue, M. Zhu, J. Liu, *Small* **2023**, *20*, 2309717.
[74] J. Song, L. Yue, Y. Dou, K. Yang, L. Zhang, L. Jiang, H. Liang, Y. Cao, T. Fan, X. Zhang, G. Qi, X. Sun, Z. Wang, *Chem. Eng. J.* **2025**, *511*, 161934.
[75] N. Wang, B. Chen, K. Qin, E. Liu, C. Shi, C. He, N. Zhao, *Nano Energy* **2019**, *60*, 332.
[76] T. Li, Y. Liu, X. Zhang, L. Zhang, Y. Wu, X. Jiang, Q. Zhou, C. Ma, L. Ni, G. Diao, *J. Energy Chem.* **2025**, *104*, 551.
[77] B. Fan, Q. He, Q. Wei, W. Liu, B. Zhou, Y. Zou, *Carbon* **2023**, *214*, 118361.
[78] M. Shi, X. Han, W. Qu, M. Jiang, Q. Li, F. Jiang, X. Xu, S. Ifuku, C. Zhang, C. Wang, J. Hu, L. Yang, Y. Lin, H. Yu, S. Liu, J. Li, Y. Wu, W. Chen, *Adv. Mater.* **2025**, *37*, 2419918.
[79] J. Gu, C. Dong, C. Zhou, C. Shen, Y. Pi, X. Xu, *Sci. China Mater.* **2023**, *66*, 2181.
[80] B. Wang, Z. Xing, S. Jia, Q. He, Y. Zhao, X. Meng, S. Tang, *ACS Nano* **2025**, *19*, 15120.
[81] H. Jia, J. Fan, P. Su, T. Guo, M. Liu, *Small* **2024**, *20*, 2311343.
[82] S. Tian, Q. Zeng, G. Liu, J. Huang, X. Sun, D. Wang, H. Yang, Z. Liu, X. Mo, Z. Wang, K. Tao, S. Peng, *Nano-Micro Lett.* **2022**, *14*, 196.
[83] X. Zuo, M. Zhen, D. Liu, L. Fu, Y. Qiu, H. Liu, Y. Zhang, *Adv. Funct. Mater.* **2024**, *34*, 2405486.
[84] Y. Zhao, Z. Shang, M. Feng, H. Zhong, Y. Du, W. Chen, Y. Wang, J. Zou, Y. Chen, H. Wang, Y. Wang, J. Zhang, G. Qu, *Adv. Mater.* **2025**, *37*, 2501869.
[85] a) J. Huang, Q. Zhang, H. Peng, X. Liu, W. Qian, F. Wei, *Energy Environ. Sci.* **2014**, *7*, 347; b) T. Zhuang, J. Huang, H. Peng, L. He, X. Cheng, C. Chen, Q. Zhang, *Small* **2015**, *12*, 381.
[86] F. Zeng, Z. Jin, K. Yuan, S. Liu, X. Cheng, A. Wang, W. Wang, Y. Yang, *J. Mater. Chem. A* **2016**, *4*, 12319.
[87] Z. Wang, W. Huang, J. Hua, Y. Wang, H. Yi, W. Zhao, Q. Zhao, H. Jia, B. Fei, *Small Methods* **2020**, *4*, 2000082.

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[88] T. Lei, W. Chen, W. Lv, J. Huang, J. Zhu, J. Chu, C. Yan, C. Wu, Y. Yan, W. He, J. Xiong, Y. Li, C. Yan, J. B. Goodenough, X. Duan, *Joule* **2018**, *2*, 2091.
[89] C. Chang, S. Chung, S. Nanda, A. Manthiram, *Mater. Today Energy* **2017**, *6*, 72.
[90] W. Ahn, S. N. Lim, D. U. Lee, K. Kim, Z. Chen, S. Yeon, *J. Mater. Chem. A* **2015**, *3*, 9461.
[91] T. Yim, S. H. Han, N. H. Park, M. S. Park, J. H. Lee, J. Shin, J. W. Choi, Y. Jung, Y. N. Jo, J. Yu, K. J. Kim, *Adv. Funct. Mater.* **2016**, *26*, 7817.
[92] Q. Li, Y. Song, R. Xu, L. Zhang, J. Gao, Z. Xia, Z. Tian, N. Wei, M. H. Rümmeli, X. Zou, J. Sun, Z. Liu, *ACS Nano* **2018**, *12*, 10240.
[93] Y. Fan, Z. Yang, W. Hua, D. Liu, T. Tao, M. M. Rahman, W. Lei, S. Huang, Y. Chen, *Adv. Energy Mater.* **2017**, *7*, 1602380.
[94] J. Sun, Y. Sun, M. Pasta, G. Zhou, Y. Li, W. Liu, F. Xiong, Y. Cui, *Adv. Mater.* **2016**, *28*, 9797.
[95] J. Balach, H. K. Singh, S. Gomoll, T. Jaumann, M. Klose, S. Oswald, M. Richter, J. Eckert, L. Giebeler, *ACS Appl. Mater. Interfaces* **2016**, *8*, 14586.
[96] P. Zeng, L. Huang, X. Zhang, R. Zhang, L. Wu, Y. Chen, *Chem. Eng. J.* **2018**, *349*, 327.
[97] Z. Zhang, Y. Lai, Z. Zhang, K. Zhang, J. Li, *Electrochim. Acta* **2014**, *129*, 55.
[98] G. Xu, Q. Yan, S. Wang, A. Kushima, P. Bai, K. Liu, X. Zhang, Z. Tang, J. Li, *Chem. Sci.* **2017**, *8*, 6619.
[99] J. Liu, L. Yuan, K. Yuan, Z. Li, Z. Hao, J. Xiang, Y. Huang, *Nanoscale* **2016**, *8*, 13638.
[100] W. Kong, L. Yan, Y. Luo, D. Wang, K. Jiang, Q. Li, S. Fan, J. Wang, *Adv. Funct. Mater.* **2017**, *27*, 1606663.
[101] X. Chen, X. Ding, C. Wang, Z. Feng, L. Xu, X. Gao, Y. Zhai, D. Wang, *Nanoscale* **2018**, *10*, 13694.
[102] M. Li, C. Wang, L. Miao, J. Xiang, T. Wang, K. Yuan, J. Chen, Y. Huang, *J. Mater. Chem. A* **2018**, *6*, 5862.
[103] Y. Luo, N. Luo, W. Kong, H. Wu, K. Wang, S. Fan, W. Duan, J. Wang, *Small* **2017**, *14*, 1702853.
[104] J. He, Y. Chen, A. Manthiram, *Energy Environ. Sci.* **2018**, *11*, 2560.
[105] W. Yao, W. Zheng, J. Xu, C. Tian, K. Han, W. Sun, S. Xiao, *ACS Nano* **2021**, *15*, 7114.
[106] Z. A. Ghazi, X. He, A. M. Khattak, N. A. Khan, B. Liang, A. Iqbal, J. Wang, H. Sin, L. Li, Z. Tang, *Adv. Mater.* **2017**, *29*, 1606817.
[107] Y. Yang, S. Wang, L. Zhang, Y. Deng, H. Xu, X. Qin, G. Chen, *Chem. Eng. J.* **2019**, *369*, 77.
[108] N. Wei, J. Cai, R. Wang, M. Wang, W. Lv, H. Ci, J. Sun, Z. Liu, *Nano Energy* **2019**, *66*, 104190.
[109] J. Park, B. Yu, J. S. Park, J. W. Choi, C. Kim, Y. Sung, J. B. Goodenough, *Adv. Energy Mater.* **2017**, *7*, 1602567.
[110] S. Yao, J. Cui, J. Huang, Z. Lu, Y. Deng, W. G. Chong, J. Wu, M. I. Ul Haq, F. Ciucci, J. Kim, *Adv. Energy Mater.* **2018**, *8*, 1800710.
[111] J. Wang, S. Yi, J. Liu, S. Sun, Y. Liu, D. Yang, K. Xi, G. Gao, A. Abdelkader, W. Yan, S. Ding, R. V. Kumar, *ACS Nano* **2020**, *14*, 9819.
[112] H. Yao, K. Yan, W. Li, G. Zheng, D. Kong, Z. W. Seh, V. K. Narasimhan, Z. Liang, Y. Cui, *Energy Environ. Sci.* **2014**, *7*, 3381.
[113] J. Huang, Z. Xu, S. Abouali, M. A. Garakani, J. Kim, *Carbon* **2016**, *99*, 624.
[114] H. J. Peng, D. W. Wang, J. Q. Huang, X. B. Cheng, Z. Yuan, F. Wei, Q. Zhang, *Adv. Sci.* **2015**, *3*, 1500268.
[115] G. Zhou, L. Li, D. Wang, X. Shan, S. Pei, F. Li, H. Cheng, *Adv. Mater.* **2014**, *27*, 641.
[116] Y. Dong, S. Zheng, J. Qin, X. Zhao, H. Shi, X. Wang, J. Chen, Z. Wu, *ACS Nano* **2018**, *12*, 2381.
[117] D. Guo, F. Ming, H. Su, Y. Wu, W. Wahyudi, M. Li, M. N. Hedhili, G. Sheng, L.-J. Li, H. N. Alshareef, Y. Li, Z. Lai, *Nano Energy* **2019**, *61*, 478.
[118] H. Shi, X. Zhao, Z.-S. Wu, Y. Dong, P. Lu, J. Chen, W. Ren, H.-M. Cheng, X. Bao, *Nano Energy* **2019**, *60*, 743.
[119] Y. Guo, S. Zhu, C. Mao, Y. Chen, L. Liu, J. Liu, X. Wang, Q. Wu, L. Yang, Z. Hu, *Adv. Mater.* **2023**, *35*, 2304551.
[120] J. Gu, C. Dong, Y. Zhu, H. Liu, J. Ji, Y. Yu, C. Ma, C. Zhou, L. Mai, X. Xu, *Angew. Chem., Int. Ed.* **2025**, *64*, 202507184.
[121] Q. Chen, S. Chen, J. Ma, S. Ding, J. Zhang, *Adv. Funct. Mater.* **2023**, *34*, 2308272.
[122] J. Feng, C. Zhang, W. Liu, S. Yu, L. Wang, T. Wang, C. Shi, X. Zhao, S. Chen, S. Chou, J. Song, *Angew. Chem., Int. Ed.* **2024**, *63*, 202407042.
[123] F. Zhang, X. Guo, P. Xiong, J. Zhang, J. Song, K. Yan, X. Gao, H. Liu, G. Wang, *Adv. Energy Mater.* **2020**, *10*, 2000446.
[124] Y. Yang, X.-J. Hong, C.-L. Song, G.-H. Li, Y.-X. Zheng, D.-D. Zhou, M. Zhang, Y.-P. Cai, H. Wang, *J. Mater. Chem. A* **2019**, *7*, 16323.
[125] L. Li, S. Basu, Y. Wang, Z. Chen, P. Hundekar, B. Wang, J. Shi, Y. Shi, S. Narayanan, N. Koratkar, *Science* **2018**, *359*, 1513.
[126] Y. Xie, H. Xiang, P. Shi, J. Guo, H. Wang, *J. Membr. Sci.* **2017**, *524*, 315.
[127] L. Ding, X. Yue, X. Zhang, Y. Chen, J. Liu, Z. Shi, Z. Wang, X. Yan, Z. Liang, *Proc. Natl. Acad. Sci. U. S. A.s* **2023**, *120*, 2314264120.
[128] H. Cui, Y. Song, D. Ren, L. Wang, X. He, *Joule* **2018**, *8*, 29.
[129] Z. Hao, C. Wang, Y. Wu, Q. Zhang, H. Xu, Y. Jin, J. Liu, H. Wang, X. He, *Adv. Energy Mater.* **2023**, *13*, 2204007.
[130] D. Guo, L. Mu, F. Lin, G. Liu, *ACS Nano* **2023**, *18*, 155.
[131] Q. Zhang, B. Hou, X. Wu, X. Li, Z. Guo, J. Liu, D. Cao, X. Huang, J. Duan, D. Mo, J. Liu, H. Yao, *Adv. Energy Mater.* **2024**, *14*, 2401377.
[132] J. Wang, Y. Liu, Q. Cai, A. Dong, D. Yang, D. Zhao, *Adv. Mater.* **2022**, *34*, 2107975.
[133] Z. Cui, D. Wang, J. Guo, Q. Nian, D. Ruan, J. Fan, J. Ma, L. Li, Q. Dong, X. Luo, Z. Wang, X. Ou, R. Cao, S. Jiao, X. Ren, *J. Am. Chem. Soc.* **2024**, *146*, 27644.
[134] L. Sheng, Q. Wang, X. Liu, H. Cui, X. Wang, Y. Xu, Z. Li, L. Wang, Z. Chen, G. Xu, J. Wang, Y. Tang, K. Amine, H. Xu, X. He, *Nat. Commun.* **2022**, *13*, 172.
[135] Y. Yang, S. Yao, Z. Liang, Y. Wen, Z. Liu, Y. Wu, J. Liu, M. Zhu, *ACS Energy Lett.* **2022**, *7*, 885.
[136] Z. Zhang, J. Wang, H. Qin, B. Zhang, H. Lin, W. Zheng, D. Wang, X. Ji, X. Ou, *ACS Nano* **2024**, *18*, 2250.
[137] S. Xu, T. Zhao, Y. Ye, T. Yang, R. Luo, L. Li, F. Wu, R. Chen, *Small* **2022**, *18*, 2104390.
[138] Y. Zhang, S. Zhang, Y. Cao, H. Wang, J. Sun, C. Liu, X. Han, S. Liu, Z. Yang, J. Sun, *Nano Lett.* **2022**, *22*, 1795.
[139] W. Lu, Y. Liu, S. Cao, P. Yi, S. He, F. Zuo, L. Ma, M. Ye, J. Shen, *Adv. Mater.* **2025**, *37*, 2500873.
[140] Y. Wang, Y. Zhang, Y. Zhang, H. Gong, S. Zhang, W. Liu, Y. Li, W. Xu, Y. Zhu, Y. Mei, J. Sun, *Small Methods* **2025**, 2500201.
[141] T. Yang, X. Xu, S. Chen, Y. Yang, F. Li, W. Fan, Y. Wu, J. Zhao, J. Liu, Y. Huo, *Angew. Chem., Int. Ed.* **2025**, *64*, 202420973.
[142] K. Wang, T. Zhao, R. Lv, W. Tang, T. Yu, G. Chen, L. Li, F. Wu, R. Chen, *Adv. Energy Mater.* **2024**, *14*, 2401281.
[143] X. Zhang, F. Ma, K. Srinivas, B. Yu, X. Chen, B. Wang, X. Wang, D. Liu, Z. Zhang, J. He, Y. Chen, *Energy Storage Mater.* **2022**, *45*, 656.
[144] L. Zuo, Q. Ma, P. Xiao, Q. Guo, W. Xie, D. Lu, X. Yun, C. Zheng, Y. Chen, *Adv. Mater.* **2024**, *36*, 2311529.
[145] J. Li, G. Yin, Y. Wang, Y. Xiang, N. Yu, X. Li, Y. Jiang, M. Xu, X. Zhang, *Chem. Eng. J.* **2025**, *507*, 160148.
[146] Z. Zhao, Y. Zhang, Y. Dou, X. Li, X. Fan, Q. Li, H. Liu, Z. Xu, B. Zhang, X. Guo, *J. Membr. Sci.* **2024**, *705*, 122885.
[147] G. Li, Y. Yang, J. Cai, T. Wen, L. Zhuang, X. Huang, Y. Cai, X. Hong, *ACS Appl. Energy Mater.* **2022**, *5*, 13554.
[148] Y. Gao, Y. Zhang, C. Han, S. Liu, S. Zhang, X. Liu, B. Zhang, F. Pna, J. Sun, *ACS Nano* **2023**, *17*, 22632.
[149] X. Cheng, R. Zhang, C. Zhao, Q. Zhang, *Chem. Rev.* **2017**, *117*, 10403.

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[150] S. Zhang, R. Li, N. Hu, T. Deng, S. Weng, Z. Wu, D. Lu, H. Zhang, J. Zhang, X. Wang, L. Chen, L. Fan, X. Fan, *Nat. Commun.* **2022**, *13*, 5431.
[151] X. He, K. Zhang, Z. Zhu, Z. Tong, X. Liang, *Chem. Soc. Rev.* **2024**, *53*, 9.
[152] Q. Zhao, R. Zhou, C. Wang, J. Kang, Q. Zhang, J. Liu, Y. Jin, H. Wang, Z. Zheng, L. Guo, *Adv. Funct. Mater.* **2022**, *32*, 2112711.
[153] S. Li, H. Yang, M. Geng, H. Do, C. Peng, *Nano Lett.* **2024**, *24*, 486.
[154] Y. Liu, Q. Liu, L. Xin, Y. Liu, F. Yang, E. A. Stach, J. Xie, *Nat. Energy* **2017**, *2*, 17083.
[155] Z. Sun, Q. Zhang, Z. Wang, Y. Chen, K. Wang, F. Shen, J. Guo, X. Han, *Energy Mater.* **2024**, *4*, 400047.
[156] J. Cannarella, X. Liu, C. Z. Leng, P. D. Sinko, G. Y. Gor, C. B. Arnold, *J. Electrochem. Soc.* **2014**, *161*, F3117.
[157] W. Zhang, Z. Tu, J. Qian, S. Choudhury, L. A. Archer, Y. Lu, *Small* **2017**, *14*, 1703001.
[158] S. Tung, S. Ho, M. Yang, R. Zhang, N. A. Kotov, *Nat. Commun.* **2015**, *6*, 6152.
[159] X. Hao, J. Zhu, X. Jiang, H. Wu, J. Qiao, W. Sun, Z. Wang, K. Sun, *Nano Lett.* **2016**, *16*, 2981.
[160] Y. Song, K. Wu, T. Zhang, L. Lu, Y. Guan, F. Zhou, X. Wang, Y. Yin, Y. Tan, F. Li, T. Tian, Y. Ni, H. Yao, S. Yu, *Adv. Mater.* **2019**, *31*, 1905711.
[161] H. Liu, C. Liu, Y. Zhou, Y. Zhang, W. Deng, G. Zou, H. Hou, X. Ji, *Energy Storage Mater.* **2024**, *71*, 103575.
[162] S. Fu, C. Hu, J. Li, H. Cui, Y. Liu, K. Liu, Y. Yang, M. Wang, *J. Energy Chem.* **2024**, *88*, 82.
[163] Z. Li, S. Zhou, X. Wu, B. Zhang, X. Yu, F. Pei, H. Liao, Y. Qiao, H. Zhou, S. Sun, *Adv. Funct. Mater.* **2023**, *33*, 2211774.
[164] Z. Chen, T. Chen, J. Wang, P. Li, J. Liu, W. Chen, Z. Yang, Y. Deng, J. Chang, Y. Yang, *Adv. Energy Mater.* **2024**, *14*, 2401568.
[165] Y. Wang, K. Zhou, L. Cui, J. Mei, S. Li, L. Li, W. Fan, L. Zhang, T. Liu, *J. Power Sources* **2024**, *591*, 233853.
[166] Y. Liu, Z. Zhang, X. Du, Y. Wang, X. Guo, M. Yu, B. Liu, W. Hu, L. Shen, Y. Lu, G. Zhu, *ACS Appl. Mater. Interfaces* **2023**, *15*, 37354.
[167] D. Chen, Y. Liu, C. Xia, Y. Han, Q. Sun, X. Wang, W. Chen, X. Jian, W. Lv, J. Ma, W. He, *InfoMat* **2021**, *4*, 12247.
[168] B. Yuan, B. Yang, P. Xu, M. Zhang, *ACS Nano* **2025**, *19*, 1981.
[169] G. Lin, K. Jia, Z. Bai, C. Liu, S. Liu, Y. Huang, X. Liu, *Adv. Funct. Mater.* **2022**, *32*, 2207969.
[170] Q. Zhang, L. Chen, X. Li, B. Hou, X. Wu, X. Gui, D. Cao, J. Liu, J. Li, J. Duan, D. Mo, J. Liu, H. Yao, *ACS Nano* **2024**, *18*, 32162.
[171] Y. Zhu, J. Xie, A. Pei, B. Liu, Y. Wu, D. Lin, J. Li, H. Wang, H. Chen, J. Xu, A. Yang, C. Wu, H. Wang, W. Chen, Y. Cui, *Nat. Commun.* **2019**, *10*, 2067.
[172] a) Z. Wang, Z. Wu, L. Weng, S. Ge, D. Jiang, M. Huang, D. M. Mulvihill, Q. Chen, Z. Guo, A. Jazzar, X. He, X. Zhang, B. Xu, *Adv. Funct. Mater.* **2023**, *33*, 2301549; b) L. Zhang, H. Deng, Q. Fu, *Compos. Commun.* **2018**, *8*, 74.
[173] P. Li, Z. Liu, Y. Peng, S. Yang, T. Meng, Y. Hu, Y. Jiang, H. Sun, Q. Li, X. Hu, *Nano Res.* **2023**, *17*, 2746.
[174] Y. Guo, Q. Wu, L. Liu, G. Li, L. Yang, X. Wang, Y. Ma, Z. Hu, *Adv. Sci.* **2022**, *9*, 2200411.
[175] W. Kang, Y. Shen, T. Yang, Z. Zhao, Y. Gou, *Adv. Funct. Mater.* **2024**, *35*, 2415432.
[176] Y. Guo, B. Feng, Y. Wang, J. Zhao, C. Zhou, X. Wang, L. Yang, Z. Jin, Z. Hu, Q. Wu, *Nano Energy* **2025**, *133*, 110472.
[177] D. Han, X. Wang, Y. Zhou, J. Zhang, Z. Liu, Z. Xiao, J. Zhou, Z. Wang, J. Zheng, Z. Jia, B. Tian, J. Xie, Z. Liu, W. Tang, *Adv. Energy Mater.* **2022**, *12*, 2201190.
[178] R. D. McGillicuddy, S. Thapa, M. B. Wenny, M. I. Gonzalez, J. A. Mason, *J. Am. Chem. Soc.* **2020**, *142*, 19170.
[179] Q. Zhang, Y. Huo, Z. Rao, *Sci. Bull.* **2016**, *61*, 391.
[180] Z. Liu, Q. Hu, S. Guo, L. Yu, X. Hu, *Adv. Mater.* **2021**, *33*, 2008088.
[181] Z. Liu, Y. Peng, T. Meng, L. Yu, S. Wang, X. Hu, *Energy Storage Mater.* **2022**, *47*, 445.
[182] G. Jiang, J. Liu, Z. Wang, J. Ma, *Adv. Funct. Mater.* **2023**, *33*, 2300629.
[183] K. Liu, W. Liu, Y. Qiu, B. Kong, Y. Sun, Z. Chen, D. Zhuo, D. Lin, Y. Cui, *Sci. Adv.* **2017**, *3*, 1601978.
[184] J. H. Yang, Y. K. Jeong, W. Kim, M. A. Lee, J. W. Choi, H. Kim, K. J. Kim, *Adv. Energy Mater.* **2024**, *15*, 2304366.
[185] P. Liu, X. Zhang, C. Ma, D. Huang, P. Li, Y. Shi, C. Qu, X. Shi, *Batteries* **2023**, *9*, 113.
[186] Z. Huang, Y. Liang, Z. Wu, Y. Kong, M. Bai, M. Li, B. Hong, T. Huang, S. Huang, H. Chen, S. Zhang, *Adv. Mater.* **2024**, *37*, 2410318.
[187] D. Xie, Z. Wang, X. Ma, Y. Feng, X. Tang, Q. Gu, Y. Deng, P. Gao, *Energy Environ. Mater.* **2024**, *7*, 12746.
[188] Z. Chang, H. Yang, A. Pan, P. He, H. Zhou, *Nat. Commun.* **2022**, *13*, 6788.
[189] H. Jia, J. Liu, B. Liu, R. Kuphal, V. Mottini, P. Monday, M. Ball, J. Li, M. Nejad, C. Fang, *Adv. Mater.* **2025**, *37*, 2419694.
[190] D. Chen, Y. Liu, C. Feng, Y. He, S. Zhou, B. Yuan, Y. Dong, H. Xie, G. Zeng, J. Han, W. He, *Electron.* **2023**, *1*, 1.
[191] G. Yu, Y. Cui, S. Lin, R. Liu, S. Liu, Y. Zhu, D. Wu, *Adv. Funct. Mater.* **2024**, *34*, 2314935.
[192] S. Choi, U. Kim, Y. Roh, C. B. Dzakpasu, J. Lim, M. Song, J. Rhee, Y. Lee, Y. M. Lee, *J. Power Sources* **2024**, *623*, 235427.
[193] Q. Zhao, R. Wang, X. Hu, Y. Wang, G. Lu, Z. Yang, Q. Liu, X. Yang, F. Pan, C. Xu, *Adv. Sci.* **2022**, *9*, 2102215.
[194] T. Zhang, J. Chen, T. Tian, B. Shen, Y. Peng, Y. Song, B. Jiang, L. Lu, H. Yao, S. Yu, *Adv. Funct. Mater.* **2019**, *29*, 1902023.
[195] T. Qin, H. Yang, Q. Li, X. Yu, H. Li, *Ind. Chem. Mater.* **2024**, *2*, 191.
[196] Y. Sun, Y. Yuan, L. Lu, X. Han, X. Kong, H. Wang, M. Ouyang, P. Gao, H. Zheng, K. Wang, *eTransportation* **2022**, *13*, 100183.
[197] a) G. Ji, D. Tang, J. Wang, Z. Liang, H. Ji, J. Ma, Z. Zhuang, S. Liu, G. Zhou, H. Cheng, *Nat. Commun.* **2024**, *15*, 4086; b) S. Natarajan, T. Mae, H. Y. Teah, H. Sakurai, S. Noda, *J. Mater. Chem. A* **2025**, *13*, 4984.
[198] O. Guselnikova, O. Semyonov, E. Sviridova, R. Gulyaev, A. Gorbunova, D. Kogolev, A. Trelin, Y. Yamauchi, R. Boukherroub, P. Postnikov, *Chem. Soc. Rev.* **2023**, *52*, 4755.

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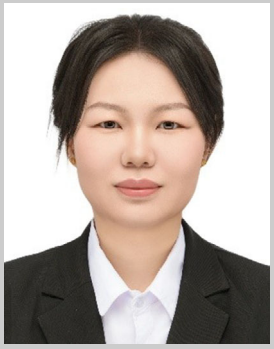
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